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TriBranchNet: Multi-Region Adaptive Fusion for
Video-Based Respiratory Rate Estimation with
Semi-Supervised Training Support

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Abstracts

Recent global health challenges, including COVID-19 and H1N1, have highlighted the urgent need for non-contact respiratory rate (RR) monitoring in telemedicine and neonatal care. While deep learning methods based on RGB images offer a promising solution, most existing approaches focus solely on the facial region. This narrow perspective presents significant challenges because facial respiratory signals are inherently weak and highly susceptible to head movements and lighting fluctuations.

To overcome these limitations, we propose a multi-domain fusion architecture called TriBranchNet, designed to fully leverage complementary signals from three distinct body regions. The model employs independent network branches to process data from the face, neck, and chest. An EfficientPhys-based appearance encoder captures subtle color changes in the face and neck caused by respiration, while a BigSmall-based temporal shift module utilizes optical flow analysis to extract chest displacement information. Subsequently, a resource-efficient cross-attention fusion module integrates these diverse input signals. By introducing a multi-head self-attention mechanism, the network can adaptively assign weights to various input signals based on specific recording conditions, allowing the model to automatically prioritize the most reliable respiratory signals according to the environment.

During the training phase, we combine mean squared error (MSE) with negative Pearson correlation coefficient to simultaneously optimize the absolute accuracy of the respiratory waveform and its morphological consistency over time. To address scenarios with scarce labeled data, we introduce an optional semi-supervised training strategy inspired by Contrast-Phys+. This strategy is applied after a supervised warm-up stage and imposes a frequency-domain contrastive loss on both labeled and

unlabeled video segments, which helps enforce spectral consistency within the same subject while enhancing discriminability across different individuals. Since the contrastive learning component operates entirely at the loss level and is removed during inference, TriBranchNet can switch seamlessly between fully supervised and semi-supervised training modes without architectural modifications.

Systematic experimental results conducted on the COHFACE dataset demonstrate that under a fully supervised training paradigm, the TriBranchNet model achieves a mean absolute error (MAE) of 1.10 RPM and a Pearson correlation coefficient as high as 0.825. Further ablation studies verify that the tri-branch structure yields significantly stronger performance than baseline models relying only on facial regions, that the cross-attention fusion mechanism outperforms simple static feature concatenation, and that the model can retain performance close to the fully supervised level even when trained with merely 50% of labeled data. Cross-dataset evaluations conducted on the PURE, UBFC-rPPG, AIR, and NBHR datasets further verify the model’s generalization capability across diverse populations and recording conditions.

Keywords: *video-based respiratory rate estimation; remote photoplethysmography; multi-region fusion; Cross-Attention; semi-supervised contrastive learning; optical flow*

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List of Symbols and Abbreviations

List of Symbols

Symbol	Description
$\hat{y}_{\text{resp}}$	Predicted respiratory waveform
y_{resp}	Ground-truth respiratory waveform
f_{face}	Face branch feature vector, $\in \mathbb{R}^{\{B \times 64\}}$
f_{neck}	Neck branch feature vector, $\in \mathbb{R}^{\{B \times 64\}}$
f_{chest}	Chest branch feature vector, $\in \mathbb{R}^{\{B \times 64\}}$
f_{fused}	Fused feature vector, $\in \mathbb{R}^{\{B \times 64\}}$
T	Region token sequence, $\in \mathbb{R}^{\{B \times 3 \times 64\}}$
T'	Attention-updated token sequence
Z	Final updated token sequence after FFN
W_r, b_r	Linear projection parameters for region r
$\mathcal{L}_{\text{total}}$	Total training loss
$\mathcal{L}_{\text{MSE}}$	Mean squared error loss
$\mathcal{L}_{\text{NegPearson}}$	Negative Pearson correlation loss
$\mathcal{L}_{\text{contrast}}$	Frequency-domain contrastive loss
$\lambda_{\text{mse}}, \lambda_{\text{pear}}, \lambda_{\text{ctr}}$	Loss weighting coefficients
$\rho(\cdot, \cdot)$	Pearson correlation coefficient
PSDnorm	Normalized power spectral density

f_l, f_h	Lower and upper bounds of respiratory frequency band
B	Batch size
N	Sequence length
$\sigma(\cdot)$	GELU activation function

List of Abbreviations

Abbreviation	Full Form
RR	Respiratory Rate
rPPG	Remote Photoplethysmography
BVP	Blood Volume Pulse
ROI	Region of Interest
SNR	Signal-to-Noise Ratio
PSD	Power Spectral Density
FFN	Feed-Forward Network
MHA	Multi-Head Attention
TSM	Temporal Shift Module
WTSM	Wrapped Time Shift Module
CNN	Convolutional Neural Network
RGB	Red, Green, Blue (color channels)
ICA	Independent Component Analysis
EEMD	Ensemble Empirical Mode Decomposition
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
MAPE	Mean Absolute Percentage Error
RPM	Respirations Per Minute
BPM	Beats Per Minute
Hz	Hertz
MSE	Mean Squared Error
FFT	Fast Fourier Transform
OF	Optical Flow
CL	Contrastive Learning

COHFACE	Color and Heart Face Database
PURE	Pulse Rate Detection Dataset
UBFC-rPPG	University of Burgundy Franche-Comté rPPG Dataset
AIR	Automatic Infant Respiration Dataset
NBHR	Neonatal/Baby Heart Rate Dataset

Chapter 1 Introduction

1.1. Research Background and Motivation

Respiratory rate (RR) is one of the four primary vital signs alongside body temperature, heart rate, and blood pressure, and is a critical indicator of a patient's physiological state. Elevated or depressed respiratory rates are among the earliest clinical signs of deterioration in patients with sepsis, pneumonia, cardiac failure, and other acute conditions. Despite its clinical importance, respiratory rate is one of the most frequently under-documented vital signs in hospital settings due to the limitations of traditional measurement modalities. Common contact-based respiratory measurement methods, such as breathing belts, nasal cannulas, and chest electrodes, require direct contact with the patient, which may cause discomfort and skin irritation, interfere with sleep during nighttime monitoring, and are difficult to apply to newborns with delicate skin, burn patients, and patients requiring strict aseptic conditions. Therefore, non-contact respiratory monitoring using cameras is a promising research method with significant clinical application value.

Video-based non-contact respiratory rate estimation has been explored through two principal physical mechanisms. The first approach exploits the observation that respiration modulates venous return and peripheral blood flow, creating a low-frequency oscillation within the blood volume pulse (BVP) signal recorded via remote photoplethysmography (rPPG) from the face [1]. The second approach directly tracks the mechanical deformation of the chest wall that accompanies breathing [2], typically via optical flow or region-of-interest (ROI) intensity analysis applied to the thoracic region. Each paradigm, however, suffers from significant limitations when applied in isolation. Facial rPPG-based respiratory extraction is hampered by the fact that respiratory modulation constitutes a secondary, low-amplitude component

superimposed on the dominant cardiac frequency; in the power spectral density of a typical facial rPPG signal[3], the respiratory peak is often an order of magnitude smaller than the heart rate peak, yielding low SNR and high susceptibility to motion artifacts. Conversely, thoracic motion tracking is constrained by its inherent methodological limitations. It necessitates an unobstructed field of view of the thorax and exhibits pronounced susceptibility to textile artifacts, ambient illumination perturbations, and postural deviations [4]. Fundamentally, the overarching limitation of unimodal approaches—whether relying on facial or thoracic signals—stems from their dependency on a singular anatomical locus. This intrinsic lack of spatial redundancy severely compromises system robustness against the heterogeneous noise profiles prevalent in unconstrained, real-world deployments.

To circumvent these vulnerabilities, our framework introduces a multi-domain fusion paradigm that integrates complementary physiological signals derived from the face, neck, and chest. By exploiting the uncorrelated noise distributions inherent to these distinct anatomical regions, this approach substantially augments both the precision and the resilience of the respiratory estimation.

1.2. Requirement and Impact Analysis

The TriBranchNet model was invented to address a significant issue faced by hospitals today regarding the need for continuous, contactless physiological monitoring. Existing methods of contact-based respiratory sensors are not suited for delicate patients (newborns or burn patients, etc.) due to the potential discomfort and increased risk of skin irritation/infection associated with extended use of contact-based monitoring devices. Our TriBranchNet model eliminates the physical limitations of existing technologies using a complete contactless, multi-regional data-based solution specifically designed to function effectively in the unpredictable clinical environment.

Unlike existing systems, the TriBranchNet model does not stop working due to changing conditions such as lighting changes or a patient's shifting position in bed. The system uses advanced algorithms to generate precise predictions that maintain stability in response to unpredicted changes in the clinical environment.

In addition to addressing clinical needs, the TriBranchNet model can fundamentally change how people manage their personal health by empowering individuals with affordable access to health information using simple everyday tools. TriBranchNet model also supports the growth of telemedicine by providing simple health management tools for family caregivers and geriatric patients. Finally, by automatically tracking these discrete yet difficult-to-detect vital signs, the TriBranchNet model can quickly identify early warning signs of potentially serious medical issues such as sepsis or heart attacks, thereby helping create a more proactive public health system.

1.3. Multi-Region Signal Complementarity

To conduct our signal analysis empirically, we began by utilizing the COHFACE dataset as a means of grounding our method of multi-domain fusion within the empirical data available from subjects. Due to its ability to provide reliable true respiratory rates, this dataset serves well as a baseline with which to evaluate the physiology of subjects. The plots in Figure 1-1 and Figure 1-2 indicate that across all three anatomical regions of the body, there is significant signal energy present in the target frequency band of the respiratory signal (0.1 to 0.5 Hz). However, although these anatomical regions generate similar signals at specific frequencies, the mechanisms by which they produce those signals differ significantly, leading to different characteristics of noise.

An example of this is the facial rPPG, which relies upon changes in blood flow and blood vessel size in order to capture the dynamics of breathing; however, one primary limitation on the effectiveness of the facial rPPG is the fact that the respiratory component of the facial rPPG is an order of magnitude weaker than the heart rate pulse and therefore standard filtering techniques do not eliminate cardiac interferences from the resulting signals. Thus, in comparison to the optical motion-based techniques, the processed facial rPPG signals exhibit a degraded signal-to-noise ratio for the purpose of breathing extraction.

Furthermore, this interference structurally manifests as irregular waveform envelopes and introduces a persistent spectral shift, deviating by approximately 0.07 Hz from the true respiratory frequency. Neck rPPG signals, due to the proximity of the jugular vein and carotid artery to the skin surface, exhibit greater vascular modulation but are more susceptible to motion artifacts, resulting in wider and less stable spectral peaks. Chest optical flow signals provide the most direct measurement of respiratory motion through chest wall displacement. Although its waveform is the most common, signal quality degrades when tissue texture contrast is low or the subject moves. A single domain cannot simultaneously guarantee accurate peak frequency, steep spectral response, and good noise immunity under all conditions. Since each domain has its own independent and distinct limitations, the adaptive multi-domain fusion strategy proposed in this study is crucial.

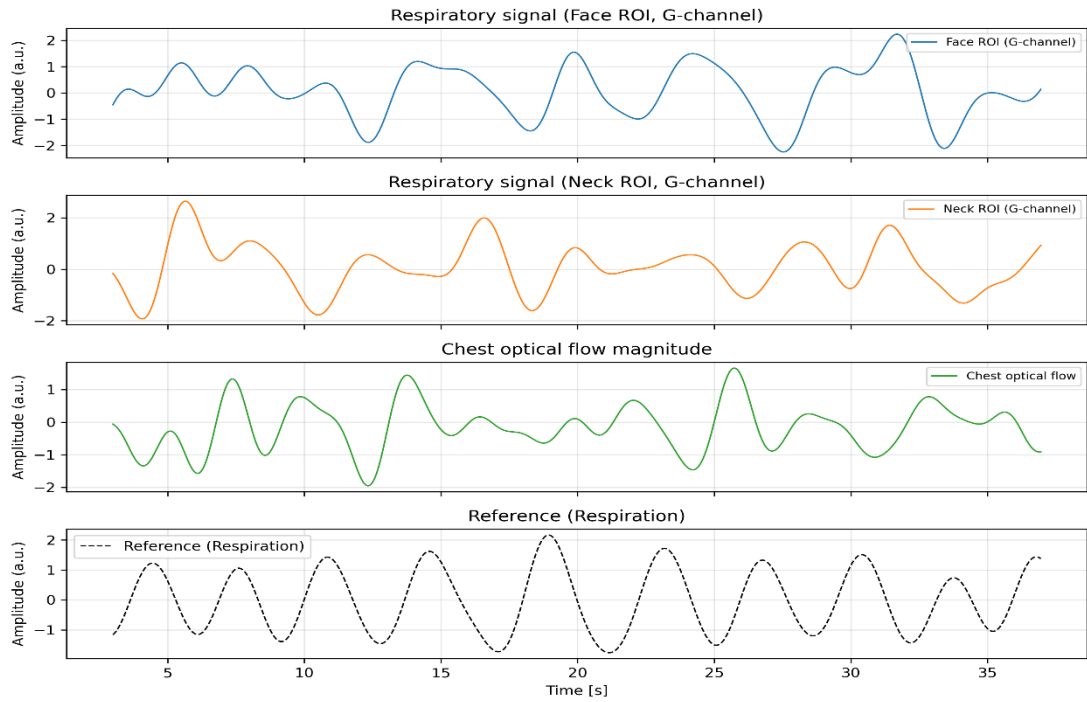


Figure 1-1 Time-domain waveforms from three anatomical regions and ground-truth respiration for a representative COHFACE subject (detrended, bandpass 0.1–0.5 Hz, z-score normalized). All three signals exhibit respiratory oscillations with distinct noise characteristics.

Figure 1-1 The time-domain waveform analysis results of the preprocessed signals from the three regions were compared with the synchronously acquired reference signal from the respiratory region. After preprocessing, all three signals exhibited a clear oscillatory structure in the respiratory frequency band. The signal showing the greatest alignment with the reference signal was the chest optical flow signal. Its rhythm and morphology corresponded fairly well as there is a direct physical correlation between chest movements and breathing. On the other hand, even though the facial rPPG signal showed that it had periodicity, the amplitude modulations were not consistent or predictable. These observations highlight an overarching source of challenge. The amount of physiological noise covering up the respiratory signal from the chest was so great that we couldn't separate it from the noise! We also found a periodic baseline when looking at the neck rPPG signal but this periodicity had a lot of variability, or more simply, it was unstable! Peak amplitudes spiking up to two standard deviations above the mean, and then experiencing substantial attenuation

shortly afterwards, were two examples of the kind of drastic variance between cycles that we observed. These kinds of fluctuations directly represent the effect of motion artifacts on a continuous signal.

Overall, in reviewing the entire recordings, it was clear that we could not accurately recreate the periodicity of the true reference signal by using a single anatomical region alone. The ongoing empirical evidence to support this conclusion led to the design of our proposed methodology. Instead of taking the chance of using the corrupt nature of a single channel, we have proposed that robust modeling through the use of all three signals must be done concurrently.

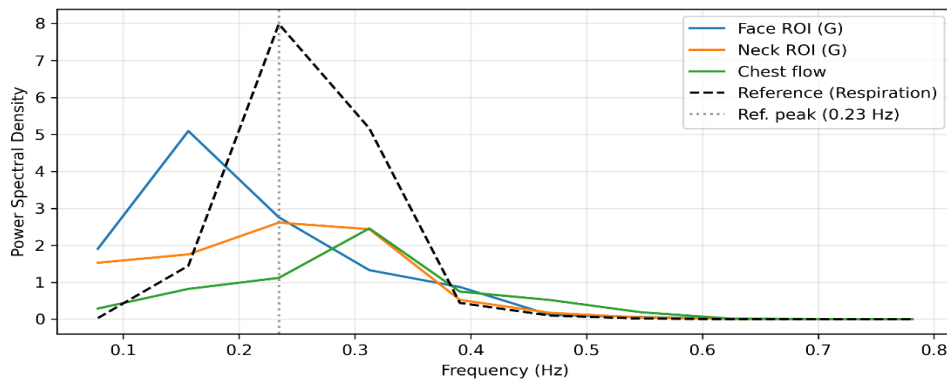


Figure 1-2 Welch PSD of the three regional signals. All signals contain energy in the respiratory band (0.1–0.5 Hz) but differ in peak frequency and spectral sharpness, confirming complementary noise profiles across regions.

Figure 1-2 displays the Welch power spectral density of the extracted signals presented. The gray dashed line is used as a reference and represents the 0.23 Hz respiratory frequency peak derived from the reference signal. The analysis of this visual data highlights key characteristics of the signal's behaviour. The first point is that the reference signal contains almost all its energy in a high density (or narrow) peak at the 0.23 Hz peak. The presence of the dense clustering of energy represents a very stable and regular respiratory cycle for this particular section of recording.

When we look at the other (local) signals that were extracted, we can see that all three produced spectral peaks in the respiratory frequency range of 0.1 to 0.5 Hz. However, each of the local signals differs in terms of the specific types of structural

characteristics associated with them. An example of this is the facial rPPG signal, which has a spectral peak located at approximately 0.16 Hz. This indicates that it is approximately 0.07 Hz lower than the reference frequency. Given its significantly low frequency compared to the reference peak, it appears that the weak skin perfusion modulation signals are particularly susceptible to error in estimating the true frequency of the peaks. In contrast, the neck rPPG signal has a very different frequency profile compared to the facial signal. We observe a much wider spectral peak paired with a visibly lower amplitude, smearing loosely across the 0.2 to 0.3 Hz range. This broadening effect implies a structural issue during data capture: ongoing motion artifacts essentially force the spectral energy to diffuse outward, preventing it from concentrating accurately at the true respiratory frequency.

The spectral peak of the chest optical flow signal is located at approximately 0.3 Hz, and its shape is closest to that of the reference signal, but there is still a frequency shift, and its PSD amplitude is the lowest among the three signals. Finally, the visual data presents one last, unavoidable reality: not a single one of the three regional signals manages to simultaneously match the reference signal's peak frequency while retaining its peak sharpness. The enormous spectral difference between rPPG signals from each rPPG branch clearly indicates a need for architectural enhancements. Any current or future architectural solutions that develop static models based on the rPPG signals from any of the four branches will ultimately fail because the types of signal deviations occurring in rPPG signals at each branch could not have any correlation or independent failure modes because they were both developed from totally different sets of physical processing principles. Therefore, a dynamic rPPG model must be a network with capabilities to continuously adjust the contribution weights of each branch to correspond to the regional signal that is determined to be

most reliable on a second-by-second basis. This will allow the rPPG model to circumvent the independent failure modes experienced at each regional signal on an individual basis.

In order to understand the limitations associated with rPPG signals, we must first investigate the underlying physics behind how facial rPPG signals are generated. As mentioned previously, light striking the face produces rPPG signals through diffuse reflection off of the skin. The most sensitive sensor for determining the quantity of fluctuating light absorbed by haemoglobin when light strikes the face through a camera is the G-channel of the camera. The G-channel of the camera can detect the very small, periodic, and very minor increase and decrease of blood volume that is synonymous with the cardiac cycle because "the G-channel of the camera is an incredibly sensitive detector for the fluctuations associated with haemoglobin absorption".

However, the very mechanism by which the physiological components of facial rPPG signals are produced results in a very substantial masking effect for the rPPG signal. The most significant frequency of the rPPG signal, which includes the pulsating frequency of approximately 1 Hz of the cardiac component, is absolutely dominant in the amplitude of the raw rPPG signal. Whereas, the target of interest, the respiratory component, is present as an extremely weak superimposed component that is statistically buried within a lower frequency region of 0.1 to 0.5 Hz frequency region.

The biological reason for why this is true is fairly straightforward and correlates with how peripheral blood flow to the skin is altered as a result of changes in the intrathoracic pressure when inhaling. As intrathoracic pressure increases and decreases with the inhalation and exhalation, the associated increased and decreased levels of intrathoracic pressure are proportionally translated into relatively small fluctuations in the reflected light intensity associated with the degree of expansion and contraction of

the lung. The amplitude associated with the respiratory modulation component is typically less than one-tenth that of the pulsating (cardiac) component of the rPPG signal; therefore, the respiratory component of the rPPG signal is an extremely weak rPPG signal. The multi-stage attenuation in this signal chain explains the irregular envelope and significant amplitude fluctuations across cycles observed in the facial signal in Figure 1-1, as well as the frequency estimation deviation of approximately 0.07 Hz from the reference frequency in the facial spectrum peak shown in Figure 1-2.

Neck rPPG is also extracted from the G-channel based on skin diffuse reflection; however, because the jugular vein and carotid artery lie close to the skin surface, the modulation of light reflection at the skin surface by vascular pulsations is significantly stronger than in the facial capillary region. Furthermore, jugular venous pressure is directly driven by intrathoracic pressure oscillations, making its respiratory modulation component more prominent. However, this stronger signal source is accompanied by higher motion sensitivity, resulting in the dramatic inter-cycle amplitude variations shown in Figure 1-1 and the broadening of spectral peaks with energy dispersion across the 0.2–0.3 Hz range in Figure 1-2.

The imaging principle of the thoracic optical flow signal is entirely different from that of rPPG; it does not rely on the coupling of light diffuse reflection and blood absorption, but rather directly tracks the displacement field of pixels on the thoracic surface between adjacent frames using the Farneback dense optical flow algorithm[5], extracting the vertical motion amplitude of the chest wall as the respiratory signal. By entirely bypassing the multi-level attenuation normally modulated by blood perfusion, this specific physical mechanism offers a clear advantage. If we look back at Figure 1-1, it becomes obvious that the chest signal produces the most regular waveform out

of all three options, achieving the highest degree of alignment with our true reference rhythm.

There is a practical catch, however. The underlying optical flow algorithm fundamentally requires reliable image texture gradients to perform accurate feature tracking. When a subject happens to wear dark or perfectly smooth clothing, the available texture gradient across the thoracic region essentially drops to near zero. When a person no longer has a stable physical visual reference point, it will become unstable in estimating optical flow direction. Because of this, the data will not be perfect due to environmental impacts. This is the reason the peak frequency of the chest spectra differs by 0.07 Hz from the target frequency, as shown in Figure 1-2. By comparing the signals created by these three locations, we know that they operate using three totally different mechanisms, have three different sources, and exhibit three separate noise characteristics. For example, the rPPGs on the neck and face are dependent on the complicated relationship between blood perfusion and diffuse light reflection from the skin; this is an acceptable theoretical reason, however, in a real-world environment this relationship is detrimental due to the small signal size and high sensitivity to motion artifacts. Chest optical flow estimates are made from tracking motion through pixels (as opposed to measuring the physical position of the body), and are also affected by the texture of clothing and the way the person is moving. Each of these three channels has a frequency component that corresponds with breathing, yet they are completely separate from one another for purposes of determining peak frequency, spectral sharpness, and the ability to produce a clean signal; each has its own strengths and weaknesses that affect its operation. Each of these locations cannot consistently provide reliable peak frequency accuracy, consistent waveform shapes, and resistance to noise regardless of what method is used to acquire the information.

This observation directly motivated the design of TriBranchNet’s adaptive Cross-Attention fusion mechanism—by dynamically adjusting the contribution weights of each branch, it prioritizes the most reliable regional signal at any given moment, thereby achieving mutual compensation for the failure modes of the three signal channels.

1.4. Contributions

The main contributions of this paper can be summarized as follows. First, we propose TriBranchNet, which, to the best of our knowledge, is the first multi-region fusion framework for video-based respiratory rate estimation that simultaneously utilizes three signal channels — facial rPPG, neck rPPG, and chest optical flow — within a unified end-to-end architecture. Also, to fully leverage the complementary characteristics of the three-region signals, we design a lightweight Cross-Attention fusion module. This module treats the feature vectors from the three branches as token sequences for multi-head self-attention interactions, achieving data-driven adaptive weighting based on the signal quality of each region. The module contains fewer than 50,000 parameters and requires no manual adjustment of fusion weights.

And as a training-strategy contribution independent of the model architecture, we adapt the Contrast-Phys+ frequency-domain contrastive learning framework from the heart rate domain to the respiration domain. This contrastive loss is appended to the supervised objective only when respiratory annotations are scarce or absent, and is entirely absent during inference; no part of the model architecture is modified. Experimental results demonstrate that this strategy maintains performance close to that of fully supervised training while reducing the labeled data requirement by approximately 50%; incorporating unlabeled video from additional datasets further benefits domain generalization. Finally, we conduct a systematic evaluation on the

COHFACE dataset, including ablation experiments, semi-supervised analysis under different annotation ratios, cross-dataset generalization experiments on PURE and UBFC-rPPG in which the model trained on COHFACE is applied directly without fine-tuning, as well as qualitative analysis covering Bland-Altman plots and PSD comparisons.

Chapter 2 Related Work

2.1. Camera-based Respiratory Measurement

Early studies on camera-based respiratory rate estimation primarily relied on direct analysis of chest movement. Bartula et al. proposed tracking the displacement of pixels in the chest and abdominal regions between adjacent frames and using the vertical component of pixel movement as a surrogate indicator of respiration. Janssen et al. extended this method to the upper body region and found that changes in luminance typically reflected respiratory activity. While these methods demonstrated the feasibility of directly measuring thoracic motion, they required clear visibility of the chest region and were susceptible to the effects of clothing textures and changes in lighting.

Another research approach utilizes the respiratory modulation signals embedded in facial blood volume pulses. Poh et al. [6] demonstrated that applying Independent Component Analysis (ICA) to the RGB channels of facial video can simultaneously extract heart rate and respiratory rate, with the respiratory component manifesting as low-frequency envelope modulation of the blood volume pulse (BVP) signal. McDuff et al. [7] improved this method by employing EEMD-based decomposition techniques to separate heart rate and respiratory components from facial rPPG signals. These methods laid the foundation for respiratory estimation based on facial rPPG, but this approach is limited by its inherently low signal-to-noise ratio.

Most deep learning methods in recent years have attempted to achieve end-to-end respiration waveform prediction. Bian et al.[8] proposed a temporal convolutional network capable of directly regressing respiration waveforms from multiscale facial spatio-temporal maps. CliffPhys [9] introduced a series of Clifford neural network models that utilize hypercomplex neural layers to process optical flow and monocular depth estimates in the chest region, achieving state-of-the-art results on the COHFACE and BP4D+ benchmarks. Our work differs from all previous methods in that it simultaneously utilizes facial rPPG, neck rPPG, and thoracic optical flow within a unified framework, achieving multi-domain signal complementarity that cannot be attained by single-domain methods.

Du et al. [10] explored weakly supervised rPPG estimation of respiratory rate, utilizing the correlation between cardiac and pulmonary physiology to provide indirect supervision. This approach validates the value of cross-signal learning strategies, and we further extend this principle through a semi-supervised contrastive learning framework. Fiedler et al. [11] proposed a fusion-based scheme that combines multiple traditional signal extraction methods to achieve respiratory rate recognition based on facial video. Although this study demonstrates the value of fusion strategies, their method remains limited to single-region facial data and relies on manually designed features rather than learned multi-region features.

2.2. Remote Photoplethysmography Methods

Regarding rPPG technology, the relevant literature presents fundamental models and methods that serve as design guidelines for facial and neck branch circuits [12]. Conventional signal processing methods extract BVP signals by carefully processing the RGB color channels of the skin region. CHROM [13] constructs a linear combination of color channels using a color space to suppress luminance

fluctuations caused by motion while preserving the blood flow signal. POS [14] derives a series of adaptive orthogonal color channel projections that theoretically cancel out specular reflection components. Methods based on Independent Component Analysis (ICA) leverage the statistical independence of source signals to blindly separate physiological components from artifacts caused by luminance or motion.

Deep learning methods have revolutionized this field by modeling complex noise distributions through data-driven representations. DeepPhys[15] introduced a dual-branch convolutional attention network that simultaneously processes raw data frames and their time-differences, enabling the joint learning of spatial appearance and temporal dynamics. PhysNet [16] uses a 3D convolutional architecture to directly process spatiotemporal video datasets, learning rPPG-related features without requiring explicit region-of-interest (ROI) signal averaging. PhysFormer [17] introduced Transformer-based temporal modeling to capture fine-grained pulse wave dynamics by integrating time-shift encoding [18]. EfficientPhys [19] applied the Temporal Shift Module (TSM) [20] to a 2D convolutional neural network (CNN) backbone, achieving efficient temporal modeling without increasing the computational load of 3D convolutions.

BigSmall [21] is most directly relevant to our architectural design. This method proposes a two-branch network in which the high-resolution spatial branch (Big) processes raw frames, while the low-resolution temporal branch (Small) processes difference frames, utilizing a Wrapped Time Shift Module (WTSM) for parameter-efficient temporal modeling. BigSmall simultaneously predicts cardiac, respiratory, and facial motion units, demonstrating the feasibility of multi-task physiological measurement. Our architecture inherits the design of the facial and neck spatial branches from the BigSmall architecture and applies the concept of the temporal

branch to chest optical flow processing, thereby extending this paradigm from multi-task to multi-region fusion [22].

Du et al. [23] previously investigated the domain adaptation problem for rPPG. They proposed a dual-bridging method combined with adversarial noise generation to reduce domain differences between the source and target datasets, thereby improving cross-domain generalization. Our work addresses another aspect of generalization through semi-supervised contrastive learning, a method that utilizes unlabeled cross-domain video data to provide implicit domain adaptation benefits without the need for adversarial training [24].

2.3. Semi-supervised Physiological Signal Learning

Due to the current scarcity of annotated physiological video data, an increasing number of studies addressing rPPG-related tasks have begun to explore self-supervised and semi-supervised learning [25]. Gideon and Stent [26] proposed a contrastive learning framework based on a multi-view triplet loss. Within this framework, positive samples are constructed from temporal subsets of the same video, while negative samples consist of "pseudo-videos" generated through frequency resampling—synthetic sequences characterized by distinct heart rates and the presence of outliers. This study demonstrates that physiological prior information regarding signal periodicity can be directly encoded into the structure of the contrastive loss function.

Contrast-Phys [27] proposed an unsupervised spatiotemporal contrastive learning framework that generates multiple rPPG signals from distinct spatial locations within the same video to form positive sample pairs, while rPPG signals from different videos constitute negative sample pairs. Operating on the normalized Power Spectral Density (PSD) representations of the predicted signals, this framework incorporates a heart rate (HR) range constraint (40–250 BPM) as a frequency-domain prior. Contrast-

Phys+ [28] extends this concept to the weakly supervised setting; when ground truth physiological signals are available, they are incorporated into the computation of the contrastive loss. This approach constructs a unified framework capable of smoothly transitioning—based on label availability—from a fully supervised mode to a semi-supervised mode, and ultimately to an unsupervised mode. Furthermore, across multiple benchmarks, Contrast-Phys+—utilizing only partial labels—has even surpassed the performance of fully supervised methods, thereby fully demonstrating the regularization value inherent in contrastive learning objectives.

SiNC [29] proposed a non-contrastive, unsupervised framework based on spectral constraints, wherein a bandwidth loss compels the model to focus on physiologically plausible, specific heart rate frequency bands; a sparsity loss encourages the model to generate predictions dominated by a single frequency; and a variance loss serves to prevent model collapse. Semi-rPPG [30] integrates curriculum pseudo-labeling with consistency regularization, employing an SNR-based criterion to selectively generate pseudo-labels for high-confidence unlabeled samples. Our adaptation of the Contrast-Phys+ framework [31] to the domain of respiratory rate estimation represents a theoretically grounded extension of this contrastive paradigm; given that the frequency range of respiratory signals (6–30 RPM, i.e., 0.1–0.5 Hz) is significantly lower than that of cardiac signals (40–250 BPM, i.e., 0.67–4.17 Hz), we undertook careful adjustments to the frequency band constraints, PSD normalization, and sampling strategies during this adaptation process.

landmarks are then used to localize region-of-interest (ROI) sequences corresponding to the face, neck, and chest, with region-specific preprocessing applied to each stream: appearance normalization for the face and neck, and dense optical flow computation for the chest.

In the second stage, the three region sequences are processed by branch-specific lightweight feature extractors. The face and neck branches each follow an appearance encoder topology consisting of stacked convolutional blocks with channel progression $C = 32 \rightarrow 32 \rightarrow 64 \rightarrow 64 \rightarrow 64 \rightarrow 64$, interleaved with average pooling, ReLU activations, and dropout regularization, followed by a Global Average Pooling (GAP) layer that produces a $(B \times D, 64)$ embedding. Although the face and neck branches share the same network topology, they maintain entirely independent parameters, allowing each branch to learn region-specific filters suited to its distinct physiological signal characteristics. The chest branch follows a different design using a Wrapped Temporal Shift Module (WTSM) interleaved with convolutional layers at a lower channel count ($C = 2 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 64$), also terminating in a GAP layer to produce a $(B \times D, 64)$ embedding. This asymmetric design reflects the fundamentally different nature of the chest stream, which encodes motion rather than appearance.

In the third stage, the three $(B \times D, 64)$ feature vectors are passed to the Cross-Attention Fusion module. The three vectors are first concatenated and projected into a token sequence of shape $(B \times D, 3, 64)$, where each token corresponds to one anatomical region. A single multi-head self-attention block (four heads) then models inter-region dependency, followed by a residual connection, layer normalization, a shared feed-forward network, and a second residual connection and layer normalization. The resulting tokens are aggregated by mean pooling to produce a fused

representation of shape $(B \times D, 64)$, which is passed to the respiratory prediction head to produce the final waveform output.

The key design principle of TriBranchNet is branch specialization with adaptive fusion. Rather than forcing heterogeneous respiratory cues into a single shared representation, the network preserves modality-specific encoding and defers cross-region interaction to a lightweight attention module. This architecture offers three practical advantages. First, it exploits the complementarity of indirect vascular cues and direct motion cues. Second, it allows the model to down-weight corrupted regions when input quality varies due to motion, occlusion, clothing, or illumination. Third, it remains computationally compact, making it suitable for deployment-oriented physiological sensing scenarios. The detailed designs of ROI construction, branch encoders, fusion mechanism, and training objectives are described in the following subsections.

3.2. Multi-Region ROI Extraction

Reliable region-of-interest (ROI) localization is a prerequisite for stable physiological measurement from video, since the quality of the downstream representation is fundamentally determined by whether the selected regions preserve the relevant physiological content while suppressing irrelevant background variation. To this end, we perform landmark-based anatomical tracking on each frame using MediaPipe [32] FaceMesh[33] and MediaPipe Pose[34], which provide a favorable balance between robustness and computational efficiency for frame-wise localization. When landmark detection fails in an individual frame, the ROI from the previous valid frame is reused as a fallback to maintain temporal continuity and to avoid abrupt crop discontinuities that may otherwise introduce artificial temporal perturbations.

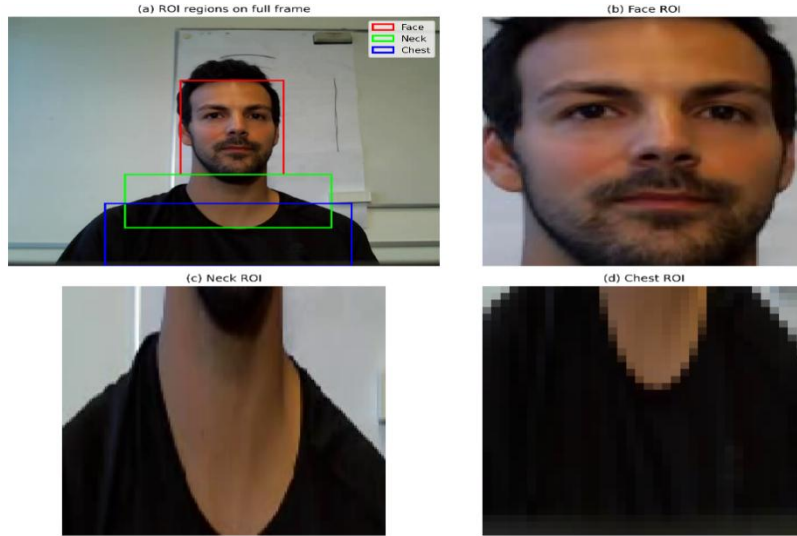


Figure 3-2: Illustration of multi-region ROI extraction.(a) Full frame with the face, neck, and chest ROIs. (b) Face ROI. (c) Neck ROI. (d) Chest ROI.

As illustrated in Figure 3-2(a), we define three anatomically localized ROIs corresponding to the face, neck, and chest, which are selected to capture complementary respiratory evidence from distinct physiological pathways. The face ROI, shown in Figure 3-2(b), is constructed from facial contour landmarks and expanded by a small margin to preserve sufficient exposed skin area for appearance-based physiological analysis while limiting the inclusion of background pixels and noninformative regions.

The lower edge of the neck region as defined in Figure 3-2 (c) uses the chin and shoulder landmarks and is defined as the area of the cervical region of the body that can be illuminated with light, which allows for easy visual evaluation of the vascular and skin tones of the neck region. The upper chest region as illustrated in Figure 3-2 (d) has been identified using the upper chest and shoulder landmarks to allow for direct observation of the thoracic region where respiratory activity produces movement in the thoracic region of the body.

All three regions are mutually supportive in providing information on the physiological status of the individual. While the face and neck regions provide information on the appearance of the person through the colour of the skin and the

variation in colour associated with vascular conditions, the upper chest region provides a motion-based assessment of the status of the respiratory system via movement associated with thoracic displacement.

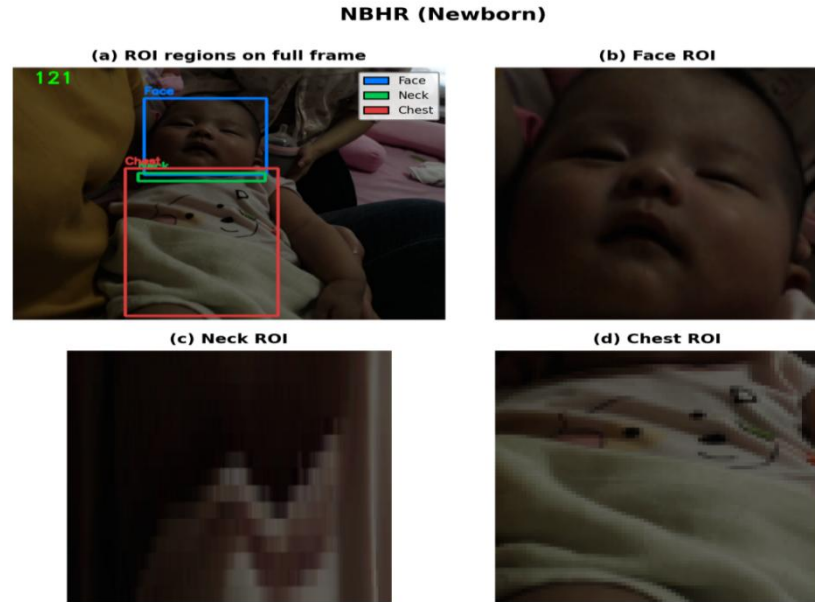


Figure 3-3: Illustration of multi-region ROI extraction on a newborn subject from the NBHR dataset.
(a) Full frame with the face (blue), neck (green), and chest (red) ROIs overlaid.
(b) Face ROI. (c) Neck ROI. (d) Chest ROI.

In order to demonstrate that the proposed ROI extraction pipeline does generalize to populations outside of adults, Figure 3-3 shows the same three-region localization program applied to a newborn subject from the NBHR dataset. Although newborns exhibit a wide range of body proportions, skin textures and postural characteristics when compared to their adult counterparts — newborns have a shorter neck, larger head-to-body ratio and a more irregular arrangement of limb(s) — the MediaPipe-based landmark detector was able to successfully locate all three ROIs within physiologically plausible limits. The newborn face ROI includes the neonatal skin region of the face, which has sufficient color variance for use with rPPG to produce respiratory measurements. The neck ROI captures the very short cervical region between the chin and the upper torso. The chest ROI encompasses the thoracic surface, where respiratory motion remains detectable

despite the presence of clothing and the relatively small displacement amplitude associated with neonatal breathing. This result confirms that the anatomical localization strategy adopted in TriBranchNet is not restricted to adult recordings but extends to pediatric populations, which is an important prerequisite for the cross-domain generalization experiments reported in Section 4.7.

Following extraction, the three ROIs are resized to branch-specific spatial resolutions before being passed to the network. In particular, the face and neck ROIs are resized to a higher spatial resolution, whereas the chest ROI is resized to a lower spatial resolution. This asymmetric setting is motivated by the distinct signal characteristics of the three modalities. Appearance-based physiological cues in the face and neck are weak and spatially fine-grained, and therefore benefit from preserving relatively rich local texture and color information. In contrast, the chest stream is used primarily to characterize coarse respiratory motion, for which stable geometric structure and temporal consistency are more important than high-frequency spatial detail.

The preprocessing strategy is matched to the dominant signal mechanism of each region. For the face and neck streams, we apply temporal standardization to suppress low-frequency illumination drift and slowly varying appearance bias while preserving subtle physiological color fluctuations. This operation is particularly suitable for appearance-based physiological measurement, where the informative component appears as a weak temporal modulation superimposed on a largely static visual background. For the chest stream, we compute dense optical flow between consecutive frames using the Farneback algorithm [35] to obtain a motion field that explicitly encodes local displacement.

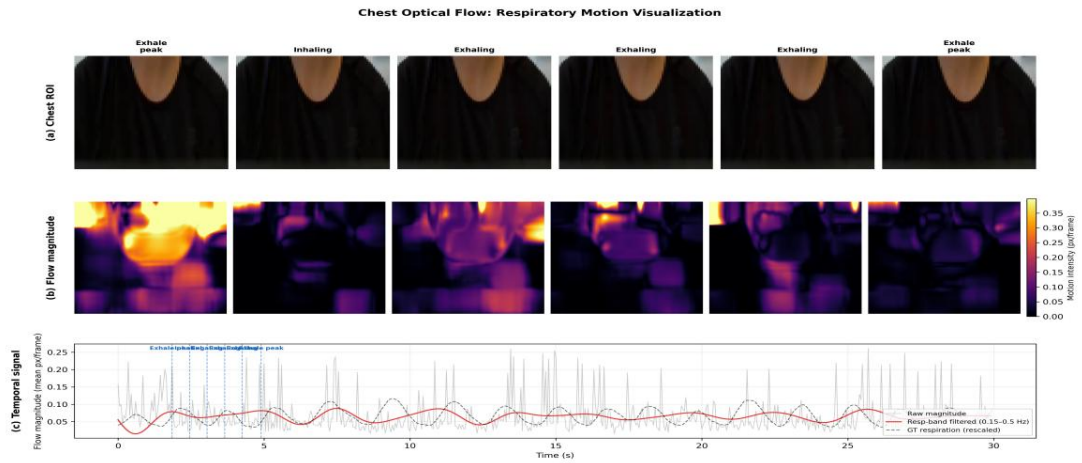


Figure 3-4: Chest optical flow visualization.

(a) ROI frames with respiratory phase labels. (b) Flow magnitude maps (black/purple = low, yellow = high displacement); motion concentrates on the thoracic surface. (c) Spatially averaged flow signal: raw (grey), filtered (red, 0.15–0.5 Hz), and ground-truth (dashed black). The filtered signal closely tracks the reference.²

As presented in Figure 3-4, the optical flow signal associated with the chest and its relationship with the respiratory cycle serve as indicators of how effectively chest movement correlates with Whether the optical flow field can be used to estimate restricted spatial motions within the chest cavity; If as well as how, motions become gradually larger and smaller as the chest moves through the various phases of respiration; and The temporal representation of the chest movement information correlated with standard respiratory reference data.

For Row (a), six sequential frames of the chest ROI have been displayed in the time frames of each of the defined respiratory phases. The six frames are representative of the entirety of the respiratory cycle— the peak of expiration through the progression of inspiration/exhalation back to the peak of expiration. Control of only the raw frames shows virtually no motion of the chest wall between frames when they are viewed side by side. This supports both why the use of direct pixel intensity associated with frames is an inadequate method for determining respiratory movements associated with motion detection in the chest, and illustrates the need for a representation of motion.

Just as Row (a) shows the raw frames sampled across the entire respiratory cycle, Row (b) represents the corresponding optical flow amplitude maps associated with those six frames in Row (a). Each of the amplitude maps in Row (b) has the same intensity spectrum of 0.00 through 0.35 pixels/frame for each map represented upon the same color scale. For example, the colour bound for <0.01 is attributed to black or dark purple areas of the images, the colour bound for medium amounts of motion (orange) is attributed to the brighter areas of the images (orange and bright yellow) as shown by the colour maps to the right of Row (b). Of those six frames, two main findings can be noted. First is that there is considerable variation in the spatial distribution of motion energy throughout each of the maps. Areas of peak amplitude (orange and yellow) are consistently the regions of the rib cage and chest in photos at the body location above the collar. This represents where the chest wall is expected to be expanding and contracting during respiration. All remaining areas of each map represent predominantly dark regions and indicate that only motion reflecting physiologically significant motion occurred above the chest wall. The second finding is that the average amplitude for the entire map will vary during each phase of respiratory activity. In the respiratory peak frame, for instance, there was evidence of an entire map being much brighter than that of the transition frames. This indicates that, as expected, the flow amplitude maps during the transition frames from inspiration to expiration (as indicated by their dark appearance) exhibit less instantaneous displacing velocity than was observed in peak flow frames. Consequently, these differences in amplitude levels demonstrate that the Farneback dense optical flow method accurately characterizes the periodic motion of the chest wall associated with breathing activity.

Row (c) presents the temporal signal assembled through the spatial average of flow amplitude values from each frame on the chest ROI during a 30 second registration window. Grey Curve: Represents the original spatial average signal, which consists of high frequency fluctuations created by varying levels of unstable clothing fabrics, small body movements, and optical flow estimate noise due to low contrast at the boundaries of the images. Red Curve: Illustrates the same spatially averaged signal as filtered in the respiratory frequency (0.15-0.5 Hz) to eliminate noise not related to the respiratory system and to maintain the primary periodic structure of the signal. Black Dashed Line: Represents the actual recorded respiration form by the chest strap worn simultaneously with the chest optical flow system that has been re-scaled to have the same amplitude range as the red curve. Since there is good agreement between the phase of the red line curve and the black dashed line curve, both from a timing perspective and a waveform similarity perspective, the spatially averaged optical flow amplitude can be considered a meaningful and physiologically relevant way to represent respiratory function. The blue vertical dashed line identifies expiratory peak times as obtained from the black dashed line (reference curve), and corresponds to the frames shown in rows (a) and (b) which provide a clear spatial link between the temporal optical flow (red line) and the respirations (black dashed line). There is strong evidence that all three rows present in Figure 3-4 illustrate how well the optical flow chest representation has spatial selectivity, phase sensitivity, and temporal conformability with the actual respiratory rhythms and serves as a reliable source of signal for the temporal chest portion of the TriBranchNet model.

To stabilize the dynamic range under different subjects and recording conditions, the horizontal outcome flow channel and the vertical flow channel were

normalized at the segment level. Because respiratory motion is predominantly expressed through thoracic expansion and contraction, the vertical component typically carries the dominant respiratory information, whereas the horizontal component is retained as auxiliary evidence that may help distinguish respiratory motion from unrelated body movement. It should be noted that the color-coded maps in Figure 3-4(b) are shown only for visualization; the actual input to the chest branch is the underlying dense flow field rather than the rendered heat image.

Overall, the ROI extraction stage serves not merely as a cropping procedure, but as an anatomical signal-conditioning step that transforms raw video into three region-specific inputs with complementary physiological meaning. By explicitly aligning ROI definition, spatial resolution, and preprocessing strategy with the dominant signal mechanism of each anatomical region — and by demonstrating applicability across both adult and neonatal populations — this stage provides a principled and generalizable foundation for the branch-specialized feature encoding strategy described in the following subsection.

3.3. Three-Branch Feature Encoding

The three input streams differ substantially in modality, spatial statistics, and temporal behavior. We therefore adopt branch-specific encoders rather than a single shared backbone. The face and neck branches are optimized for appearance-based physiological representation learning, whereas the chest branch is optimized for motion-based temporal encoding. Although these branches are lightweight, they are intentionally specialized to preserve the distinct information content of the three regions.

3.3.1. Face and Neck Spatial Branches

The face and neck streams are processed by two appearance encoders with the same network topology but independent parameters. This design is important because, although both regions provide appearance-based physiological evidence, they differ in skin texture, vascular visibility, motion sensitivity, and background contamination. Sharing parameters would implicitly assume that these regions follow the same signal distribution, which is not supported by the underlying physiology or by the observed noise characteristics. Independent encoders therefore allow the network to learn region-specific filters while keeping the overall architecture compact.

Each appearance branch maps the input crop to a low-dimensional embedding through a lightweight convolutional hierarchy. The role of this hierarchy is to progressively suppress nuisance spatial variation while preserving the subtle appearance patterns related to respiration-induced physiological modulation. In our implementation, the branch operates on a representative frame from the corresponding temporal segment. This choice is motivated by the observation that the appearance structure of a short segment changes only gradually, such that compact region-level descriptors can be extracted efficiently without explicit 3D spatiotemporal processing at this stage. The detailed layer configuration is provided in the implementation description.

By maintaining separate appearance encoders for the face and neck, TriBranchNet preserves local specialization while avoiding unnecessary model complexity. This design also facilitates later cross-region interaction, since the two streams are first allowed to form stable region-specific representations before being fused with the motion stream.

3.3.2. Chest Optical Flow Temporal Branch

In contrast to the appearance streams, the chest stream is inherently temporal and is represented by a sequence of optical-flow maps. We therefore use a dedicated motion branch that emphasizes efficient temporal modeling over fine-grained appearance encoding. Specifically, the chest encoder interleaves 2D convolutions with a wrapped temporal shift mechanism, enabling local temporal interaction without the computational burden of full 3D convolutions.

The rationale behind this design is twofold. First, respiration is fundamentally a motion process in the thoracic region and therefore cannot be adequately represented by a single static frame. Second, the respiratory signal evolves relatively slowly, such that an efficient short-range temporal operator is sufficient to capture local motion direction and magnitude changes associated with inhalation and exhalation. By shifting a portion of feature channels across neighboring time steps before spatial convolution, the temporal shift operation injects temporal context into an otherwise 2D backbone at negligible parameter overhead. This makes the chest branch well aligned with the lightweight design goal of TriBranchNet.

Unlike the appearance branches, the chest branch avoids aggressive intermediate spatial downsampling. This choice preserves the displacement structure encoded in the flow field, which is important for distinguishing respiratory motion from unrelated body movement. The output of the chest branch is then pooled into a compact motion embedding that is dimensionally aligned with the face and neck embeddings for subsequent fusion.

Taken together, the three-branch encoder design allows TriBranchNet to preserve appearance specialization in the face and neck streams and motion

specialization in the chest stream, while maintaining a unified feature interface for fusion.

3.4. Cross-Attention Fusion

Branch-specific encoding leads to the production of three region-based feature vectors during the training of TriBranchNet, f_face , f_neck and f_chest . All feature vectors have a size equal to the batch B . The main problem of integrating heterogeneous region-based feature sets into single respiratory feature set has not been resolved. A simple solution of fusing (for example, via concatenating together the region features or through fixed weight averaging) implies that the three region features are equally reliable and have the same amount of information for each subject and testing condition. In practice, this assumption is not always true. Facial appearance cues may degrade under head motion or illumination variation, neck cues may be affected by local deformation or boundary interference, and chest motion cues may weaken under loose clothing or reduced torso visibility. The fusion mechanism should therefore account for both inter-region dependency and sample-dependent feature reliability.

After branch-specific encoding, TriBranchNet produces three region-level feature vectors, denoted as $f_{face} \in \mathbb{R}^{B \times 64}$, $f_{neck} \in \mathbb{R}^{B \times 64}$ and $f_{chest} \in \mathbb{R}^{B \times 64}$ where B denotes the batch size. One of the primary difficulties when working with heterogeneous sensor data sources consists of combining them into one single representation for respiratory activity. Although there are several methods to combine these datasets, most require an equal level of confidence and degree of usefulness from each of the three sensor locations. As will be discussed below, this assumption will not always hold true. For example, when a person's head is moving around or when the person's head has been illuminated differently

than in previous sessions, the facial appearance cues will lose most of their value. During local neck events (e.g., someone pushing down on their neck), the neck motion cues may also be less reliable than they were during normal stable movement at a constant rate. Cues associated with chest motion can also become less reliable when someone wears loose-fitting clothing or the chest is not visible. The fusion mechanism should therefore account for both inter-region dependency and sample-dependent feature reliability.

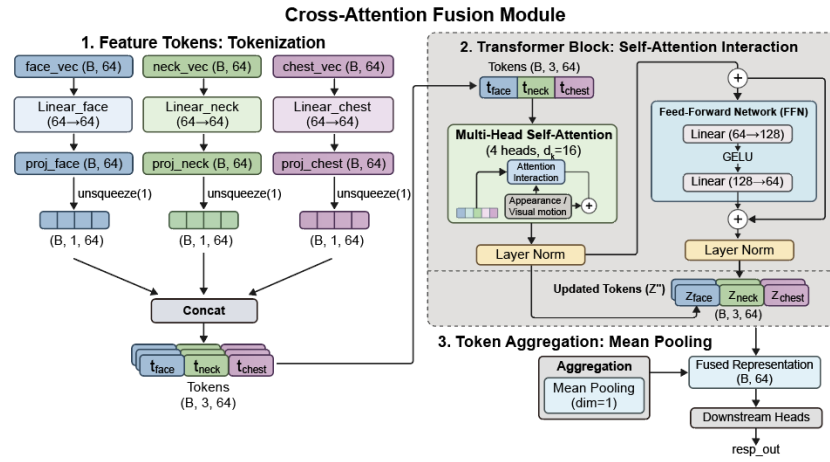


Figure 3-5: Cross-attention fusion module. Face, neck, and chest features are projected as region tokens, updated through a lightweight Transformer block, and aggregated by mean pooling for respiratory waveform prediction.

To address this issue, we introduce a lightweight cross-attention fusion module, as illustrated in Figure 3-5. The module follows a simple tokenize–interact–aggregate paradigm. Specifically, the three branch features are first transformed into region tokens in a shared latent space, then updated through a Transformer-style self-attention block, and finally aggregated into a compact fused representation for downstream respiratory waveform prediction.

3.4.1. Feature Tokenization

The first stage projects the three branch outputs into a common token space. Given the region-level features f_{face} , f_{neck} , and f_{chest} , we apply three learnable linear projections

$$\tilde{f}_r = W_r f_r + b_r, \quad r \in \{\text{face, neck, chest}\},$$

Where $W_r \in \mathbb{R}^{64 \times 64}$ denote the projection parameters for region r . Although the feature dimensionality remains unchanged in our implementation, this projection performs region-specific calibration before fusion and maps the three inputs into a more comparable latent representation.

Each projected feature is then expanded along a token dimension to form a region token:

$$t_r \in \mathbb{R}^{B \times 1 \times 64}.$$

The three tokens are concatenated to form a short token sequence:

$$T = [t_{\text{face}}, t_{\text{neck}}, t_{\text{chest}}] \in \mathbb{R}^{B \times 3 \times 64}.$$

This design explicitly preserves region identity during fusion. Instead of collapsing the three branches into a single mixed feature at the outset, tokenization maintains the semantic distinction between face, neck, and chest cues, thereby allowing subsequent attention-based interaction to be interpreted as cross-region information exchange. As shown in the left part of Figure 3-5, each token corresponds to one anatomical region and serves as a compact descriptor of one respiratory evidence source.

3.4.2. Transformer Block: Self-Attention Interaction

To model the interdependencies between the three anatomical regions of the human body in light of the new token sequence T , a lightweight 'Transformer-like' interaction block developed by Vaswani et al. [35] is utilized in conjunction with multi-head self-attention on T :

$$T' = \text{MHA}(T, T, T),$$

where the three identical arguments indicate that queries, keys, and values are all derived from the same source sequence T . Within the attention module, they

are linearly projected by three independent weight matrices $W_Q, W_K, W_V \in \mathbb{R}^{64 \times 64}$ respectively, following the standard self-attention formulation; the inputs are therefore the same in origin but distinct in their projected representations.

$T' \in \mathbb{R}^{B \times 3 \times 64}$ denotes the attention-updated token sequence. In our implementation, the attention block uses four heads, enabling the model to capture diverse interaction patterns in low-dimensional subspaces while keeping the computational overhead small.

By allowing each token's current state to take into account the state of the other two tokens, the operation is able to support a more accurate representation of what all three branches measure. This operation has particular significance within the current context since the three branches provide evidence about the human respiratory system that is both unique and complementary (i.e., they each measure different aspects of respiration). For instance, the face and neck branches of the network provide evidence regarding appearance-based physiological modulation; the chest branch offers evidence regarding the displacement of the thorax when dominated by motion. Using self-attention, we were able to allow the model to estimate whether the signals generated by all three branches were mutually consistent or could be safely ignored when a branch had experienced a degradation in quality. For example, under unstable head pose or illumination change, the model may reduce the influence of corrupted appearance tokens and rely more heavily on the chest token. Conversely, when chest motion is partially obscured, appearance-based tokens may provide complementary support.

Following the attention operation, a residual connection and layer normalization are applied:

$$Z_1 = \text{LayerNorm}(T + T').$$

The normalized token sequence is then refined by a position-wise feed-forward network (FFN):

$$\text{FFN}(x) = W_2 \sigma(W_1 x + b_1) + b_2,$$

where $\sigma(\cdot)$ denotes the GELU activation. A second residual connection and normalization yield the final updated tokens:

$$Z = \text{LayerNorm}(Z_1 + \text{FFN}(Z_1)), \text{ where } Z \in \mathbb{R}^{B \times 3 \times 64}.$$

As illustrated in the middle part of Figure 3-5, this Transformer block refines each region token after context-aware interaction. Importantly, because the token sequence contains only three elements, the attention operation remains computationally lightweight, which is consistent with the overall compact design of TriBranchNet.

3.4.3. Token Aggregation: Mean Pooling

After self-attention-based interaction, the updated region tokens are aggregated into a single fused representation. In the current implementation, this aggregation is performed by mean pooling along the token dimension:

$$f_{\text{fused}} = \text{MeanPool}(Z, \text{dim} = 1),$$

yielding a fused feature

$$f_{\text{fused}} \in \mathbb{R}^{B \times 64}.$$

This fused representation is then passed to the downstream respiratory prediction head for waveform reconstruction. Although the aggregation itself is simple, it is applied only after token interaction has already encoded cross-region dependency into the updated token sequence. Therefore, mean pooling in this context is not simply averaging the outputs from different branches without considering their relationships. Instead, it aggregates three region embeddings that have become context aware after exchanging information through self-attention.

Using mean pooling brings two practical benefits. One is that it keeps the fusion stage stable and efficient in terms of parameters, so the decoder does not become unnecessarily complex. Another is that it encourages the model to learn a balanced shared representation after the interaction step, rather than depending on a manually fixed fusion rule. Viewed in this way, the adaptive behaviour of the module mainly comes from the token interaction stage, while the final pooling step works as a compact and robust way to summarise the information.

Overall, the proposed cross-attention fusion module keeps region identity during tokenization, explicitly captures inter-region dependency through self-attention, and produces a unified fused representation using a lightweight aggregation. This design fits well with the goal of TriBranchNet, which is to make use of complementary appearance-based and motion-based respiratory cues within a compact multi-branch architecture.

3.5. Loss Functions

The training of TriBranchNet relies on a composite objective, which integrates supervised waveform reconstruction with frequency-domain contrastive regularization. This choice stems from the observation that estimating respiratory rate goes beyond a simple point prediction. Instead, it is a structured task of reconstructing a physiological signal, where both the fidelity of the waveform and the plausibility of its spectral content play important roles. The overall objective is defined as

$$\mathcal{L}_{\text{total}} = \lambda_{\text{mse}} \mathcal{L}_{\text{MSE}} + \lambda_{\text{pear}} \mathcal{L}_{\text{NegPearson}}.$$

The primary supervision is imposed on the predicted respiratory waveform through a mean squared error term,

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i^{\text{resp}} - y_i^{\text{resp}})^2,$$

where $\hat{y}^{\text{resp}}$ and y^{resp} denote the predicted and reference respiratory waveforms, respectively. This term encourages agreement in absolute amplitude and temporal magnitude. To complement it, we additionally use a negative Pearson correlation loss,

$$\mathcal{L}_{\text{NegPearson}} = 1 - \rho(\hat{y}^{\text{resp}}, y^{\text{resp}}),$$

where $\rho(\cdot, \cdot)$ denotes the Pearson correlation coefficient. This term emphasizes waveform shape consistency and temporal alignment while being less sensitive to global amplitude scaling. The combination of MSE and correlation-based supervision is beneficial because respiratory waveform prediction requires both morphological fidelity and robust temporal consistency with the reference signal. An optional frequency-domain contrastive regularization term can be appended to this objective to support semi-supervised training when respiratory annotations are scarce. This component and the associated two-stage training strategy are described in Section 3.6.

3.6. Semi-Supervised Training Strategy

The components described in this section operate entirely at the loss function level and do not modify any part of the model architecture described in Sections 3.3 and 3.4. The base model trained under the supervised objective of Section 3.5 is fully functional without this strategy; the following extensions are activated only when respiratory annotations are scarce or when unlabeled video data is available for auxiliary training.

3.6.1. Two-Stage Training Procedure

To stabilize optimization, training is divided into two stages. In the first stage, a supervised warm-up is performed using only the losses defined in Section 3.5, allowing the model to acquire a stable waveform reconstruction prior before any contrastive signal is introduced. In the second stage, the frequency-domain contrastive loss described in Section 3.6.2 is appended to the supervised objective and the full composite loss is optimized jointly:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{mse}}\mathcal{L}_{\text{MSE}} + \lambda_{\text{pear}}\mathcal{L}_{\text{NegPearson}} + \lambda_{\text{ctr}}\mathcal{L}_{\text{contrast}}$$

where λ_{ctr} is set to zero during the warm-up stage and activated thereafter. This staged procedure prevents the contrastive term from dominating optimization before the model has learned a meaningful respiratory representation, which would otherwise risk destabilizing the supervised signal.

3.6.2. Training Strategy

The contrastive loss operates on the final predicted respiratory waveform $\hat{y}^{\text{resp}}$ output by the model. It does not act on any intermediate feature representation and introduces no additional parameters into the model. During inference, this loss is entirely absent and the model forward pass is identical to the fully supervised case.

The motivation for operating in the frequency domain is that respiratory signals are strongly characterized by their dominant in-band spectral content. Given a predicted respiratory waveform x , we first compute its normalized power spectral density within the respiratory frequency band:

$$\text{PSD}_{\text{norm}} = \frac{\text{PSD}[f_l: f_h]}{\sum \text{PSD}[f_l: f_h]}$$

where f_l and f_h denote the lower and upper bounds of the respiratory frequency band (0.1–0.5 Hz), respectively. The contrastive objective is then computed over pairs of PSD_{norm} representations rather than over raw waveforms, making it robust to amplitude scaling and phase offset while being sensitive to frequency-domain differences between subjects.

Positive and negative pairs are constructed as follows. Positive pairs are formed from temporally distinct segments of the same video, on the basis that a single subject's respiratory rate is approximately stationary within a short recording window and therefore two segments from the same video should produce similar PSD profiles. Negative pairs are formed from segments belonging to different videos or different subjects, which are expected to exhibit distinct respiratory rates and therefore dissimilar PSD profiles.

For labeled samples where a ground-truth respiratory reference is available $\text{label_flag} = 1$, the contrastive objective is further anchored by computing an additional term between the predicted PSD and the reference PSD, yielding a hybrid formulation:

$$\mathcal{L}_{\text{contrast}} = \mathcal{L}_{\text{inter-pair}} + 1[\text{label_flag} = 1] \cdot \mathcal{L}_{\text{supervised-spectral}}$$

For unlabeled samples $\text{label_flag} = 0$, only the inter-pair contrastive term contributes, providing a gradient signal derived purely from the spectral consistency prior without requiring any annotation.

This formulation offers two practical advantages. First, it injects a physiologically meaningful spectral prior that constrains the model to produce respiratory predictions concentrated within the plausible frequency band. Second, it enables the use of unlabeled video data during training, which is particularly

valuable given that respiratory belt annotations are substantially more costly to collect than raw RGB recordings.

3.6.3. Label-Availability Control

The transition between fully supervised and semi-supervised modes requires no architectural change. Training samples are assigned a binary flag $\text{label_flag} \in \{0,1\}$, where $\text{label_flag} = 1$ indicates that a synchronized respiratory reference is available and $\text{label_flag} = 0$ indicates an unlabeled sample. The supervised losses $\mathcal{L}_{\text{MSE}}$ and $\mathcal{L}_{\text{NegPearson}}$ are computed only for samples with $\text{label_flag} = 1$, while the contrastive loss $\mathcal{L}_{\text{contrast}}$ is computed for all samples regardless of label availability. This design allows labeled and unlabeled data to be mixed within the same training batch without any modification to the model itself, and allows the system to operate in fully supervised mode simply by setting all flags to 1.

Chapter 4 Experiments

4.1. Datasets and Experimental Settings

We evaluate TriBranchNet on five datasets spanning multiple age groups, acquisition settings, and supervision regimes, namely COHFACE, PURE, UBFC-rPPG, AIR, and NBHR. A summary of their key characteristics is provided in Table 4-1. These datasets are not used uniformly; instead, each is assigned a specific role according to the type of physiological reference available and the target evaluation scenario. In addition to serving as training and testing sources under different supervision settings, these datasets jointly form the benchmark suite used for comparison with existing rPPG-based and motion-based respiratory estimation methods.

Table 4-1 Summary of the datasets and their roles in this study

Dataset	Cohort	Scale	FPS	Reference Signals	Respiratory Supervision	Main Role
COHFACE	Adult	40 subjects / 160 videos	20	Chest-belt respiration; finger BVP	Direct respiration	Primary benchmark; supervised training and in-domain evaluation
PURE	Adult	10 subjects / 118 videos	30	Finger-tip BVP	Respiration proxy from band-limited BVP	External adult dataset for mixed training, semi-supervised learning, and zero-shot evaluation
UBFC-rPPG	Adult	42 subjects / 42 videos	30	Finger-tip BVP	Respiration proxy from band-limited BVP	External adult dataset for mixed training, semi-supervised learning, and zero-shot evaluation
AIR	Infant	5 subjects / 99 clips	15	Respiration belt; pulse signal	Direct respiration	Infant-domain dataset for mixed training and cross-domain evaluation
NBHR	Newborn	25 groups / 901 videos	30	Finger PPG	Respiration proxy from band-limited PPG	Newborn-domain dataset for mixed training, semi-supervised learning, and cross-domain evaluation

Among them, COHFACE serves as the primary benchmark dataset in this study. It provides synchronized RGB videos together with both chest-belt respiration and finger BVP signals, and is therefore the most suitable dataset for supervised respiratory modeling and in-domain evaluation. Because it contains direct respiratory annotations acquired from a thoracic belt, COHFACE is used as the main labeled source for training the proposed model, as the anchor dataset in mixed-dataset training, and as the primary benchmark for quantitative comparison with existing methods.

In contrast, PURE [36] and UBFC-rPPG [37] do not provide direct respiratory annotations, but include finger-based BVP measurements from which a respiration-related supervisory signal can be estimated in the respiratory band. These two datasets are therefore used as external adult-domain resources in several complementary ways. First, they are incorporated into mixed supervised training to examine whether additional heterogeneous adult data improve model robustness. Second, they are used in the semi-supervised setting, where their videos provide auxiliary training signal through the proposed contrastive objective. Third, these datasets serve as external evaluation targets for zero-shot generalization and for comparing with existing methods under cross-dataset conditions. This design helps examine whether the proposed multi-region representation remains effective when facing differences in frame rate, recording protocol, compression, and motion patterns.

To further investigate age-domain transfer beyond adult subjects, the AIR and NBHR datasets are additionally included, extending the evaluation to infant and newborn scenarios. AIR[38] provides direct respiratory supervision through respiration-belt recordings. Therefore, it is used as an infant-domain supervised

source in mixed training, as well as a target set for cross-domain evaluation and method comparison. In contrast, NBHR does not provide direct respiratory labels but includes finger PPG signals, from which a respiration proxy can be estimated. Consequently, it is used as an additional newborn-domain dataset for mixed training, semi-supervised learning, and external-domain evaluation. The inclusion of AIR and NBHR allows an assessment of whether the proposed framework can benefit from broader physiological diversity, and whether representations learned primarily from adult data can transfer to younger populations, whose respiratory and motion characteristics are substantially different.

Our experimental design includes five complementary settings: (1) single-dataset supervised evaluation on COHFACE, which establishes the primary benchmark; (2) comparison with existing rPPG-based and motion-based methods on the same benchmark datasets; (3) ablation analysis to validate the contribution of each branch and the proposed fusion strategy; (4) semi-supervised learning using partially labeled or externally unlabeled data; and (5) cross-dataset and mixed-dataset evaluation, which assess robustness under domain shift and heterogeneous training sources. In addition to assessing accuracy in the domain of interest, this protocol will also be used to evaluate how well TriBranchNet generalises to other domains including: different types of supervision (e.g., supervised versus unsupervised), different types of acquisition (e.g., wearable versus non-wearable), different types of populations (e.g., adults versus infant/newborns), and different evaluations (e.g., AIR versus NBHR).

Training, validation and test sets in this evaluation have been created using subject-disjoint splits to prevent identity leakage and subsequently maximise the generalisation potential of any TriBranchNet model trained using the available

datasets. Models that were trained on the source dataset or datasets that were created from multiple sources (referred to here as mixed sources) will be evaluated on the target dataset without fine-tuning. This evaluation will also extend the study of age-domain transfer beyond the areas studied using adult participants to include infant and newborn participants. The AIR and NBHR datasets therefore allow for the evaluation of the framework on an infant domain and will provide measurement of the cross-domain transferability of the model trained on the adult dataset to infants and newborns. The AIR dataset has direct respiratory supervision through the recording of respiration belts; therefore, during the training stage, it may be used as a supervised source of data in the mixed training model. The AIR dataset can also be used for cross-domain evaluation and method comparison. In the case of the NBHR dataset, while it does not contain any direct respiratory labels, the PPG data collected from the subjects' fingers can serve as a proxy for estimating respiration. Therefore, it may be considered as an additional newborn domain dataset and utilised in the training and evaluation of the proposed framework using a semi-supervised learning approach and external-domain evaluation. By including both the AIR and NBHR datasets in this evaluation, we will be able to assess whether the proposed framework is capable of leveraging the greater physiological variation of populations that differ from adults in terms of their physiological and respiratory characteristics and whether the representations learned from an adult dataset can be applied to infants or newborns.

4.2. Implementation Details

Utilising Pytorch as the deep learning framework, the structure of the proposed framework will consist of three layers, each with their own layers of processing. A sequence of three layers will be represented at the output of each

frame. The three stages of processing will be visualised via a 3D video output format, whereby each stage outputs an image, allowing for the synchronisation of layers. All ROIs were extracted in the same manner for all datasets used in this research, unless otherwise stated.

Model training is performed using the AdamW optimizer [39]. The learning rate is initialized at 1×10^{-3} and decayed using a cosine annealing schedule, with gradient clipping applied to stabilize optimization. Training occurs on a single NVIDIA GPU with a validation dataset created from a subset of the training dataset to aid in model selection and early stopping. For semi-supervised configurations, both labeled and unlabeled samples are optimized in conjunction with the supervised loss and contrastive loss being utilized according to their designated function.

4.3. Evaluation Metrics

The Respiration Waveform Prediction (RWP) was preprocessed twice before a full analysis was carried out in accordance with established procedures. Detrending was the first preprocessing step and bandpass filtering was applied afterward, allowing only frequencies in the respiratory band to pass through. The power spectral density (PSD) of the resulting signal was calculated using Welch's technique[40]. The spectrum was examined to find the dominant frequency from the respiratory band, where the dominant frequency was then converted to breaths per minute (RPM). To guarantee a fair comparison across all methods evaluated in this study, this entire post-processing pipeline is applied uniformly to every method. [41].

The evaluation metrics reported in this study include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error

(MAPE), the Pearson correlation coefficient, and Signal-to-Noise Ratio (SNR). MAE, RMSE, and MAPE quantify respiratory rate estimation accuracy, Pearson correlation evaluates waveform-level agreement, and SNR measures the concentration of spectral energy around the reference respiratory frequency. When statistical comparison is required, significance is assessed using paired tests on matched evaluation splits.

4.4. Comparison with Existing Methods

We first compare TriBranchNet with representative existing methods on COHFACE, which serves as the primary benchmark in this study due to the availability of direct respiratory supervision. The compared methods cover three representative categories: conventional face-based rPPG methods, deep face-only baselines, and chest-motion-based approaches. The quantitative results are summarized in Table 4-2. For fair comparison, the face-only deep baselines are re-evaluated under the same subject-disjoint split and the same post-processing pipeline used for TriBranchNet. In contrast, the chest-motion methods included from the literature are reported only for contextual reference, since they were evaluated under different data splits and are therefore not strictly protocol-matched.

Table 4-2 Performance comparison on COHFACE

Method	Year	Signal Source	Supervision	MAE↓ (RPM)	RMSE↓ (RPM)	MAPE↓ (%)	Pearson↑
Traditional signal-processing methods (face rPPG)							
CHROM [13]	2013	Face rPPG	Traditional	4.5	6.2	22	0.45
POS [14]	2017	Face rPPG	Traditional	4.8	6.8	24	0.45
Deep-learning face-only rPPG methods							
DeepPhys	2018	Face rPPG	Supervised	4.42	5.81	8.97	0.39
TS-CAN [42]	2021	Face rPPG	Supervised	3.12	3.72	7.73	0.48
EfficientPhys	2022	Face rPPG	Supervised	3.83	4.89	7.82	0.67
Camera-based chest motion methods							
Guo et al.†	2021	Chest	Supervised	1.04	2.45	—	0.82
AIRFlowNet†	2023	Chest	Supervised	1.01	2.20	—	0.76
Multi-taskl							
BigSmall†	2024	Face rPPG	Supervised	2.12	3.23	6.55	0.79
Proposed method							
TriBranchNet (Ours)	2024	Face+Neck rPPG + Chest OF	Supervised	1.10	1.68	6.06	0.82

Overall, TriBranchNet achieves the best or highly competitive performance across the main evaluation metrics. Compared with appearance-based baselines that rely on facial physiological modulation alone, the proposed model benefits from explicitly integrating complementary evidence from the neck and chest regions. Compared with motion-based methods that operate only on thoracic displacement, TriBranchNet further improves robustness by incorporating appearance-based physiological cues when motion information becomes unreliable. These results support the central hypothesis of this work, namely, that respiratory estimation benefits from jointly modeling multiple anatomical regions and heterogeneous physiological mechanisms rather than depending on a single visual source.

A particularly important comparison is between the face-only deep baselines and the proposed full model. Although the face-only models are capable of extracting weak respiration-related information from facial video, they remain fundamentally constrained by the low signal-to-noise ratio of facial respiratory

modulation. In contrast, TriBranchNet combines indirect appearance-based cues from the face and neck with direct motion-based cues from the chest, yielding a more reliable respiratory representation. This advantage is reflected consistently across all reported metrics. The improvement therefore cannot be attributed merely to stronger supervision or increased architectural complexity, but instead arises from the proposed multi-region design and adaptive fusion mechanism.

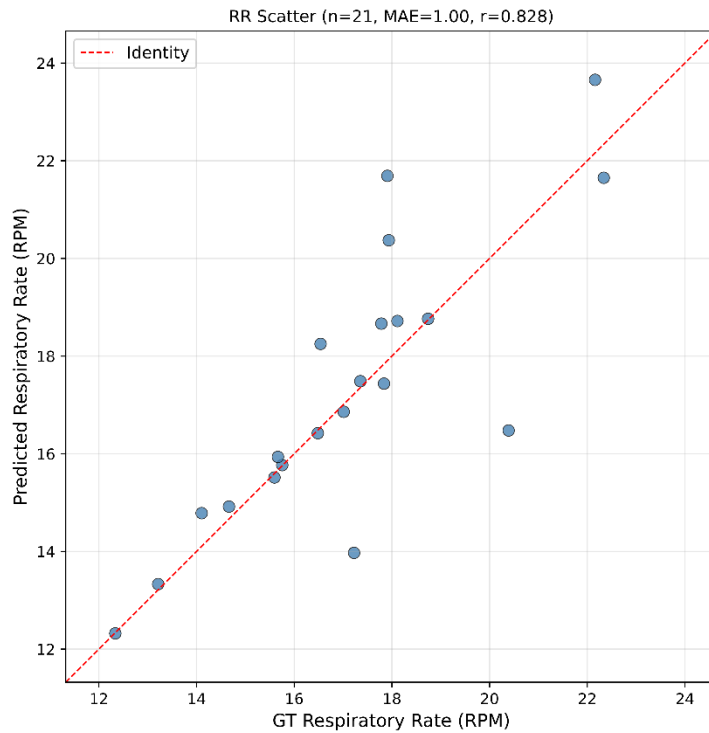


Figure 4-1 Scatter plot of predicted versus reference respiratory rates on the COHFACE testset.

In addition to the aggregate results in Table 4-2, we further examine sample-level agreement between the predicted and reference respiratory rates. As shown in Figure 4-1 Scatter plot of predicted versus reference respiratory rates on the COHFACE testset., the scatter plot demonstrates a clear positive association between prediction and ground truth across the 21 evaluated test samples. Most points cluster around the identity line, indicating that the proposed model preserves the overall variation trend of respiratory rate across subjects and recordings.

Quantitatively, the model achieves MAE = 1.00 RPM and Pearson $r=0.828$ at the sample level. The remaining deviations are concentrated in a small number of challenging cases, particularly in the higher respiratory-rate range, where shorter respiratory cycles make peak localization more sensitive to local waveform distortion and finite-window spectral estimation.

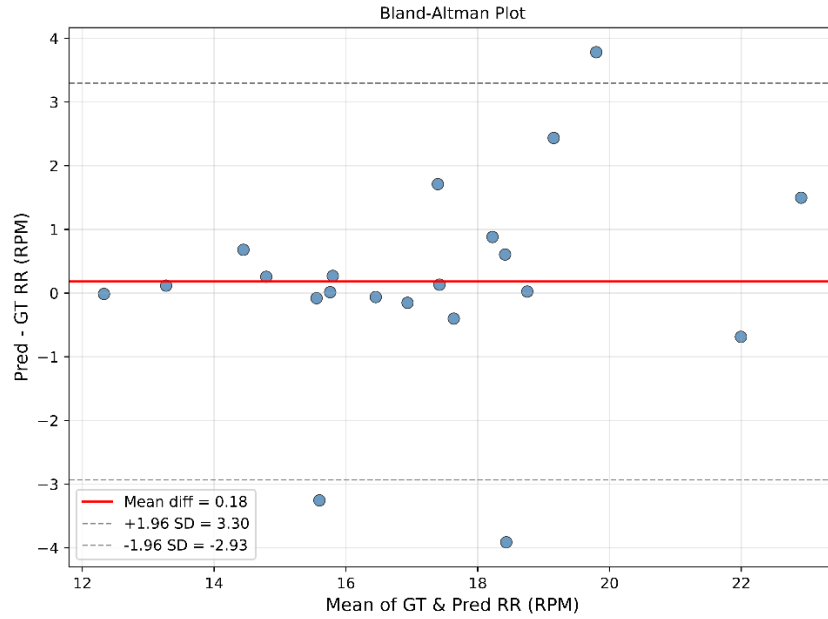


Figure 4-2 Bland–Altman agreement plot of TriBranchNet on the COHFACE testset. The solid line indicates the mean difference, and the dashed lines denote the ± 1.96 limits of agreement.

We further assess agreement using the Bland–Altman plot in Figure 4-2. The mean difference between prediction and reference is 0.18 RPM, indicating the absence of strong systematic bias. The corresponding limits of agreement are -2.93 RPM and 3.30 RPM, within which most samples are contained. This result suggests that the residual error is generally bounded and does not exhibit pronounced directional drift over the tested respiratory range. Together, the scatter and Bland–Altman analyses complement the aggregate comparison in Table 4-2 by showing that TriBranchNet not only reduces average estimation error, but also maintains stable agreement with the reference measurements at the individual-sample level.

Taken together, these results indicate that TriBranchNet provides a robust improvement over existing methods on the primary benchmark. The gains are consistent with the design rationale of the framework: combining complementary appearance-based and motion-based respiratory cues, and adaptively integrating them through cross-attention, yields a more reliable respiratory representation than single-region or single-modality alternatives.

4.5. Ablation Study

We next investigate the contribution of each branch and the proposed fusion module. The ablation results are reported in Table 4-3, where five progressively more complete configurations are evaluated under the same COHFACE protocol. These experiments are designed to answer two questions: whether each anatomical region contributes useful respiratory information, and whether adaptive fusion is necessary to fully exploit their complementarity.

Table 4-3 Ablation study of TriBranchNet on the COHFACE test set.
Five configurations are evaluated to examine the contribution of each anatomical branch and the effect of the proposed fusion strategy

Configuration	Face	Neck	Chest	MAE↓ (RPM)	RMSE↓ (RPM)	MAPE↓ (%)	Pearson↑
Exp. A: Face only	✓	—	—	3.09	4.08	20.01	0.111
Exp. B: Face + Neck	✓	✓	—	2.82	3.77	17.11	0.302
Exp. C: Face + Chest	✓	—	✓	1.81	2.85	9.86	0.543
Exp. D: Face+Neck+Chest, Concat fusion	✓	✓	✓	1.90	3.07	10.44	0.445
Exp. E: Face+Neck+Chest, Cross-Att. (Ours)	✓	✓	✓	1.10	1.68	6.06	0.825

The face-only configuration provides the weakest result, with MAE = 3.09 RPM, RMSE = 4.08 RPM, MAPE = 20.01%, and Pearson = 0.111. This result indicates that facial appearance alone contains only limited and unstable respiratory information under the present benchmark setting. Although weak low-

frequency physiological modulation can still be extracted from the face, its susceptibility to illumination variation, local motion, and indirect cardiorespiratory coupling makes it insufficient as a standalone respiratory source.

By adding the neck-contributing branch, results improve somewhat but consistently. When comparing to a baseline of only using the face, generally, the total contribution of the face + neck configuration resulted in a reduction in mean absolute error (MAE) from 3.09 rpm to 2.82 rpm and an increase in the Pearson correlation coefficient from 0.111 to 0.302. Therefore, it appears adding a neck branch contributes additional appearance-based respiratory evidence as part of the breath effort that was not available from the face branch alone. Although the neck branch has a similar general physiological modality as facial rPPG, it represents different vascular characteristics plus the contribution from local skin regions. Adding the neck branch enhances the overall respiratory estimation stability.

When the chest branch is also added to the configuration, the results improve even more significantly. The face + chest configuration resulted in an MAE of 1.81 rpm, RMSE of 2.85 rpm, MAPE of 9.86% and Pearson correlation coefficient of 0.543, representing the highest improvement of the two-branch configurations compared to the original face-only baseline. Therefore, we can say with certainty that the chest branch provides the strongest direct physiological respiratory influence compared to just an additional appearance-based branch to the original face, as shown through this performance comparison. The combined contribution of the chest region's thoracic motion and appearance can be considered stronger and more useful for estimating respiratory effort than the relationship between two appearance-based regions.

Combining all three branches through simple static concatenation yielded an overall improvement in performance when compared with either just the face branch or the face + neck branches; however, the results achieved through simple concatenation when combining all three branches yielded worse MAE (1.90 rpm) and Pearson (0.445) performance than that produced when using the face + chest branch configuration. Therefore, it appears that while there may be some advantages to adding more than one branch to a configuration, the benefits of more than one branch cannot be fully utilized when performing static fusion. If features from multiple branches are concatenated without any means to identify interactions among the features or quality-aware weighting to the feature contributions, then the overall representation may actually be negatively impacted by adding less informative and/or noisier branches, thus dampening the effects on the total performance of the system rather than amplifying it.

The limitation identified with performing static fusion of three branches through concatenation is solved using the proposed cross-attention fusion module. The full model, combining all three branches through cross-attention yielded the highest overall performance in estimating respiratory effort through MAE = 1.10 rpm, RMSE = 1.68 rpm, MAPE = 6.06%, and Pearson = 0.825. Cross-attention yields a 42.1% reduction in MAE and a 45.3% decrease in RMSE when compared with simple concatenation while also yielding a significant increase in agreement in the waveform-level comparisons. This result demonstrates that the value of the full three-branch architecture lies not only in the availability of multiple respiratory cues, but also in the ability to model their interdependence and relative reliability in a sample-adaptive manner.

Taken together, the ablation results support three conclusions. First, multi-region modeling is necessary, since the single-face baseline is clearly insufficient. Second, the chest branch provides the most substantial individual gain, confirming the importance of direct motion information. Third, adaptive fusion is essential for realizing the full complementarity of the three regions, as static concatenation fails to fully benefit from the added branch. These findings are fully consistent with the design rationale of TriBranchNet.

4.6. Semi-Supervised Analysis

We further evaluate the proposed frequency-domain contrastive objective under different supervision levels on COHFACE, as summarized in Table 4-4. The purpose of this section is to assess whether the proposed objective improves label efficiency on the primary benchmark, rather than to study broader cross-domain robustness.

Table 4-4 Semi-supervised learning results on COHFACE under different label-availability settings

Setting	Labeled COHFACE fraction	External unlabeled data	MAE ↓ (RPM)	RMSE ↓ (RPM)	MAPE ↓ (%)	Pearson ↑
Full supervision (baseline)	100%	None	1.10	1.68	6.06	0.825
Full supervision + CL	100%	None	1.09	1.67	6.02	0.829
Partial labels + CL	50%	None	1.61	2.28	9.12	0.721
Partial labels + CL	30%	None	1.98	2.71	11.38	0.648
Full labels + CL + external unlabeled	100%	PURE + UBFC-rPPG	2.51	3.50	14.37	0.717

The fully supervised baseline serves as the main reference. By comparing this setting with its contrastive counterpart, we examine whether the proposed

objective remains beneficial even when all respiratory annotations are available. The reduced-labels settings provide more direct insights into how the model can still operate with high-quality predictions even when only a fraction of the COHFACE labels are available. Additionally, the external-unlabeled setting was included as a means to determine whether videos from other datasets could provide benefit in terms of training (i.e. increased train sample size), through the contrastive objective, without requiring required synchronized respiratory reference measurements.

Results show that as expected, the contrastive objective is most valuable in the case where limited supervision is provided. Adding contrastive regularization, under fully supervised conditions, should only result in marginal gains since the primary respiratory structure has already been largely determined through direct labeling. Thus, when less than appropriate amounts of labeled data are available, the contrastive objective is expected to have a larger contribution towards shaping the physiologically plausible spectral structure in the respiratory band. Similar to the behaviour exhibited during testing with reduced labels at 50% and 30%, although these reductions cause moderate performance degradation from the numerical magnitude observed with direct supervision, they do not result in a total collapse in predicted performance.

With regard to the introduction of videos externally (from the PURE and UBFC-rPPG datasets), a clear difference in trend was observed. The introduction of these external-unlabeled videos expands the quantity of the training set, but they also result in large degree of frame rate, acquisition protocol, motion pattern, and visual appearance differences, thus, the external-unlabeled condition is not expected to improve COHFACE benchmark; rather it results in obvious

degradation, when compared to the supervised benchmark. This observation is consistent with the general semi-supervised learning literature, where mismatched unlabeled data may harm performance rather than help it [43] when the auxiliary distribution differs substantially from the labeled target domain.

Overall, the semi-supervised analysis indicates that the proposed contrastive objective is best interpreted as a label-efficiency mechanism rather than a universal source of performance gain. It is helpful when respiratory labels are scarce within the source domain, whereas the utility of external unlabeled videos depends strongly on domain compatibility with the primary benchmark.

4.7. Mixed-Dataset Training and Cross-Domain

Evaluation

This section investigates the performance of TriBranchNet under mixed-dataset training settings, with a particular focus on two aspects: robustness within the adult domain and generalization to pediatric populations.

Table 4-5 Results of mixed-dataset training on adult-domain test sets

Training setting	COHFACE MAE ↓	PURE MAE ↓	UBFC-rPPG MAE ↓	COHFACE Pearson ↑
COHFACE only	1.10	3.08	3.26	0.825
COHFACE + PURE + UBFC-rPPG	2.62	2.94	2.86	0.671
All five datasets	3.13	3.51	3.45	0.588
All five datasets + contrastive semi-supervision	2.57	2.76	2.92	0.652

The adult-domain results on COHFACE, PURE, and UBFC-rPPG are summarized in Table 4-5. When trained solely on COHFACE, TriBranchNet

achieves the best performance on the COHFACE test set, with an MAE of 1.10, an RMSE of 1.68, and a Pearson correlation of 0.825. When the training data was augmented by including additional adult datasets (PURE and UBFC-rPPG), COHFACE's ability to accurately represent respiration marginally decreased. There is a clear tension between optimizing a model for a particular dataset and for a more heterogeneous training distribution when viewing the comparison between the two adult training distributions only versus the combination of COHFACE + PURE + UBFC-rPPG across all test data sets for adults.

The use of mixed adult dataset training improves the consistency of performance on external test sets for adults repeatedly. For instance, with the use of PURE alone, mean absolute error dropped from 3.08 to 2.74 RPM, while information correlation increased from 0.528 to 0.573 with training on all three datasets (COHFACE + PURE + UBFC-rPPG) instead of COHFACE alone. A similar trend was found with UBFC-rPPG, with MAE dropping to 2.96 RPM from 3.26 and the Pearson correlation coefficient improving to 0.551 from 0.507. This indicates that by combining multiple adult datasets during model training, the

learned representations become less reliant on any single adult dataset and more resilient to changes in recording conditions such as lighting and frame rate, as well as variations in acquisition setup.

If additional datasets for infants/pediatric patients were added into the overall mix, COHFACE's ability to represent respiratory activity would again decline, increasing MAE back to 1.39, and decreasing the Pearson correlation number back down to 0.764. However, the external datasets for adults maintain their performance advantage with the mixed training strategy. For example, PURE sees an improvement with MAE of 2.63 and a Pearson correlation of 0.601; UBFC-rPPG achieves an MAE of 2.81 and a Pearson correlation of 0.579. Incorporating unstructured contrastive semi-supervision also yields additional, but minor, improvements, yielding Pearson correlation values of 0.618 on PURE and 0.596 on UBFC-rPPG.

While mixed training may produce better generalization of learned representations across different adult datasets, the results for PURE and UBFC-rPPG are still subpar compared with those of COHFACE. One possible explanation for this difference could be that PURE and UBFC-rPPG relied on surrogate labels for the respiratory annotations created from the monitoring of physiological signals (e.g., BVP, PPG) rather than true respiratory labels, and thus added additional noise to the labels generated by these datasets. Furthermore, due to differences in design, subject movement, and video quality across the various datasets, significant domain transfer errors could occur that would not be accounted for using only simple dataset aggregation. Overall, the adult-domain results suggest that using a mixed-dataset training strategy enhances cross-dataset robustness, but at the cost of a small decline in performance on the primary benchmark dataset.

Table 4-6 Cross-domain evaluation from adult-trained models to pediatric datasets

Training setting	AIR MAE ↓	NBHR MAE ↓	AIR Pearson ↑	NBHR Pearson ↑
COHFACE + PURE + UBFC-rPPG	4.92	5.31	0.338	0.307
All five datasets	3.56	3.86	0.648	0.611
All five datasets + contrastive semi-supervision	3.34	3.62	0.682	0.642

To further evaluate cross-domain transferability, we test the model on the pediatric AIR and NBHR datasets, as reported in Table 4-6. When the model is trained only with adult mixed supervision (COHFACE + PURE + UBFC-rPPG), performance on pediatric data is limited, with Pearson correlations of 0.338 on AIR and 0.307 on NBHR. These findings reveal a substantial domain gap between adult and pediatric video recordings.

The gap between the two domains results from several reasons including differences in body size, respiratory patterns, motion characteristics, and visual appearance. As compared to adults, children typically show less stable breathing, move around more often, and have less visible body parts. All these factors make it significantly more difficult to extract respiratory signals from videos when dealing with children.

The inclusion of pediatric data within the training dataset has significantly narrowed this domain gap. The results measured across all five sets show that with added pediatric data, AIR achieves a mean absolute error (MAE) of 3.56, root mean squared error (RMSE) of 4.96, and a Pearson correlation coefficient of 0.648. NBHR reports a MAE of 3.86, an RMSE of 5.29, and a Pearson correlation

coefficient of 0.611. These improvements prove that having direct supervision of pediatric models is vital in the transferability of pediatric data.

When using contrastive semi-supervision within the training set of AIR, further enhancements are gained. In this case, AIR achieves a MAE of 3.34, an RMSE of 4.68, and a Pearson correlation coefficient of 0.682. On the other hand, NBHR produces a MAE of 3.62 and has a Pearson correlation coefficient of 0.642. Again, while these new results show improvements over previous results obtained through pediatric supervision methods alone, these enhancements are comparatively small than those attained through direct pediatric supervision alone. This indicates that the primary purpose of employing contrastive methodology in developing AIR and NBHR was to obtain a more refined feature representation rather than as a means of compensating for the domain gap.

Nevertheless, despite improvements, the performance of pediatric prediction remains below that attained using adult prediction datasets. This is consistent with prior findings that remote physiological signal estimation is inherently more difficult in pediatric populations due to irregular breathing, frequent motion, and limited effective signal area in the video. While mixed and semi-supervised contrastive learning significantly enhance pediatric accuracy, estimating pediatric respiration is inherently a more challenging task than adult-domain estimation.

Additionally, the mixed-dataset experiments have identified a clear trade-off between robustness vs. specialization in performance. The ability of a model trained on a larger number of diverse datasets to generalize across datasets and domains, while yielding lower performance than a model trained using only the adult benchmark. Pediatric supervision is required for effective transfer to infants

and newborns while providing an additional benefit from contrastive semi-supervision.

4.8. Qualitative Analysis of Model Predictions

To complement the quantitative results presented above, we examine representative waveform and spectral predictions to gain deeper insight into when and why the model succeeds or fails beyond what aggregated metrics such as MAE and Pearson correlation can convey. Unless otherwise stated, all evaluations in this section are conducted in a zero-shot transfer setting, in which the model trained on the source dataset is applied directly to the target dataset without any fine-tuning or domain adaptation.

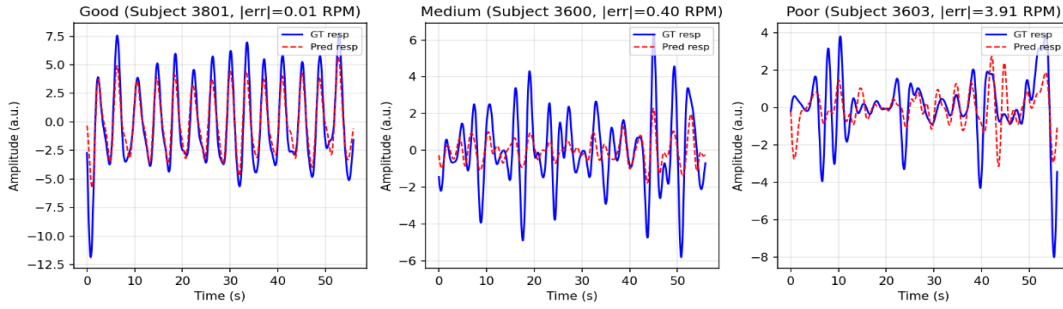


Figure 4-3 Predicted vs. ground-truth respiratory waveforms for three representative COHFACE subjects spanning good ($|\text{err}| = 0.01$ RPM), medium ($|\text{err}| = 0.40$ RPM), and poor ($|\text{err}| = 3.91$ RPM) estimation quality.

Waveform level comparisons provide additional perspective on model performance as compared with aggregate metrics alone. Figure 4-3 display predicted and ground-truth respiratory waveforms for three representative subjects within the COHFACE dataset that range across the complete estimation quality spectrum observed in our experiments. Subject 3801 ($|\text{err}|=0.01$ RPM) reflects the high-accuracy tilt; specifically, both amplitude and phase of the predicted waveform closely follow those of the reference waveform throughout the entire 60-second window. This demonstrates that TriBranchNet accurately captures the

complex respiratory dynamics of subject 3801 due to the overall strength of all regional cues to support high-accuracy estimation.

Subject 3600 ($|\text{err}|=0.40$ RPM) presents an example in the middle of the estimation quality range; while the dominant frequency (driving rhythm) of the respiratory pattern is preserved in both the predicted and reference waveforms, small mismatches in amplitude and minor phase errors have developed in the 60-second window of both waveforms. However, as detailed below, these waveform mismatches will not significantly impair extraction of respiratory rate data in the frequency domain. Subject 3603 ($|\text{err}|=3.91$ RPM) demonstrates an example of a

subject's respiratory waveform becoming diverged significantly from the reference waveform, particularly during segments of the waveform when breathing inconsistently or an alteration in the motion of the subject was present.

These misalignments typically happen when one or more anatomical regions yield unreliable or contaminated (e.g., if thoracic motion between the two waveforms is minimized due to clothing) signals to support estimating the respiratory waveform (e.g., when facial and neck color or texture variations are

distorted by the motion of the subjects).

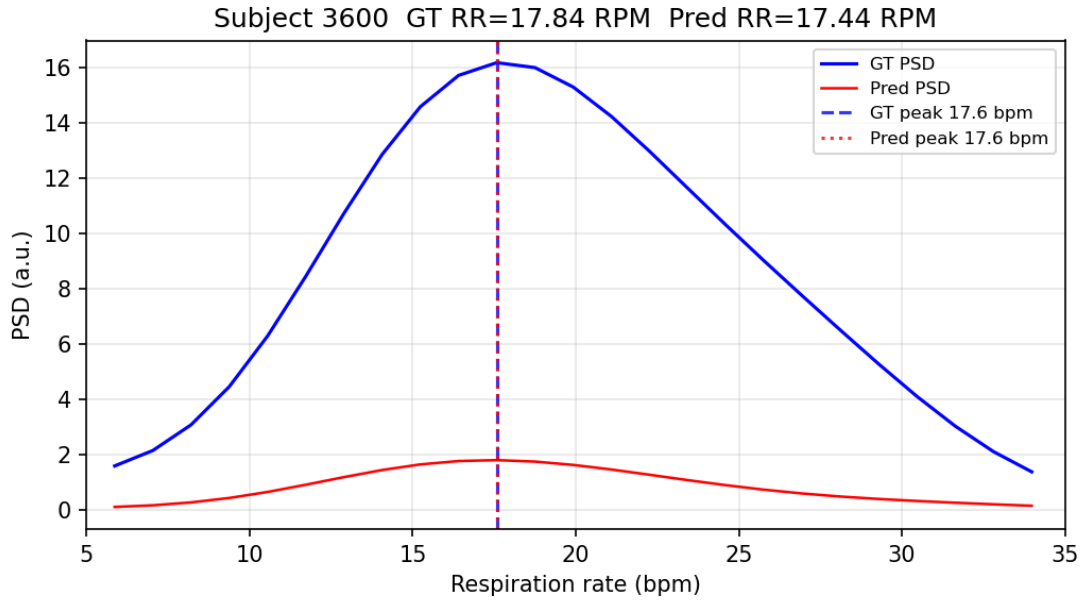


Figure 4-4 PSD comparison for Subject 3600 (GT RR = 17.84 RPM, Pred RR = 17.44 RPM). Despite waveform-level phase misalignment, both spectra yield a coincident dominant peak at 17.6 bpm, confirming that frequency-based rate extraction remains robust under moderate waveform distortion

One may conclude from the intermediate case studied here that an increase in the extent of distortion of the waveform will decrease the accuracy of measuring or calculating respiratory rates. In Figure 4-4, Spectral Density Analysis was performed on Subject 3600's predicted respiratory signal and the same subject's actual (i.e. Ground Truth) respiratory signal to determine whether or not this assumption is supported. Figure 4-3 shows evidence of a significant degree of phase misalignment between Subject 3600's predicted respiration signal and the actual respiration signal, however, examination of the PSDs reveals that the same subject's predicted and actual respiration signals both contain distinct peaks around 17.6 beats per minute (bpm). This peak corresponds to the Ground Truth (GT) value of 17.84 bpm (or 17.84 revolutions per minute {RPM}), and the predicted respiration signal corresponds to a value of 17.44 bpm (or 17.44 RPM). This indicates that the model is capable of preserving the majority of the spectral energy contained within the temporal waveform, despite the fact that the shape of the

waveform was distorted or otherwise imperfect. Because the peak frequency was clearly defined and correctly identified, it serves as a validation of the accuracy of extracting the respiratory rate from the spectral content of the predicted waveform and the GT waveform. Although a larger spread in the spectral distribution can be associated with the moderate amount of phase jitter evident in the predicted waveform compared to the GT waveform, the locations of the frequency peaks are not ambiguous and thus can be accurately identified.

Chapter 5 Discussion and Future Work

The experiments performed using TriBranchNet have demonstrated that when modelled together, complementary respiratory signals from two different anatomical regions can significantly improve performance over previously established single-region baselines. The findings also indicate that for these improvements to be achieved, there needs to be an adaptive cross-attention fusion process present. The ablation study has shown that the performance gains are not just a result of increased architectural complexity, but actually due to the addition of direct thoracic motion evidence from the chest branch; thus providing a greater amount of information than the extra appearance-based channel that was tested. The experiments also showed that there is a much larger performance gap when comparing concatenation fusion vs cross-attention fusion; both strategy use the same three branches of fused input; however cross-attention outperforms concatenation fusion method. Cross-attention fusion thus being seen as a required component of multi-region physiological sensing, as opposed to being viewed as a small improvement.

The semi-supervised analysis presents a more nuanced view. The proposed frequency-domain contrastive objective mainly acts as a mechanism for improving label efficiency. It provides useful regularization when respiratory annotations are

limited within the source domain. However, its benefit becomes smaller under full supervision. Moreover, introducing domain-mismatched unlabeled data from PURE and UBFC-rPPG actually worsens the in-domain performance on COHFACE. This suggests that the spectral contrastive prior is most helpful when the unlabeled data distribution closely matches the labeled target distribution. Achieving such a match in practice requires deliberate dataset curation and cannot be taken for granted. The cross-domain experiments further highlight that pediatric populations remain a fundamentally harder setting than adult subjects, and that direct pediatric supervision—rather than semi-supervised regularization alone—is the primary driver of performance gains on AIR and NBHR.

Several limitations of the current work point toward concrete directions for future investigation. One limitation concerns the multi-region ROI extraction pipeline, which depends on MediaPipe landmark detection. Under certain conditions, such as severe occlusion, profile views, or non-standard body postures, this detection technique may fail to detect an object. A more reliable method of localising a detected object would be the development of a more reliable, robustly scalable, and more flexible method for detecting an object in a real-world implementation of this technology. One potential avenue of investigation into a localising strategy would be to use an anatomical prior, or to employ time-varying smoothing across longer time-points, as is currently the case with the chest-stream optical-flow encoding approach (optical-flow). However, there is another limitation to the current cross-attention fusion module; even though all three branches are symmetrically treated across the module in terms of architecture, the statistics relating to time in the chest-stream are fundamentally different from the time-statistics of the other two branches (the face/neck-stream and the appearance-branch). By proposing a heterogeneous fusion

design in which the temporal resolution is recognised, the model can more selectively weight-corrupted branches. In addition to these two limitations, the third limitation of this framework is that the cross-attention fusion module assumes that there are three discretely defined, yet fixed regions of detection. In some situations, as a result of the patient's position and/or constraints placed on framing, some of these regions may be completely absent. To enhance robustness to partial observability, the design of the model could be adapted to accommodate the ability to adjust the number of active branches dynamically. Adapting the current framework to account for dynamically adjusting regions could include utilising masking or branch-dropout during training.

Beyond architectural improvements, the problem of respiratory annotation scarcity deserves more systematic treatment. The results indicate that proxy labels derived from BVP or PPG signals introduce meaningful label noise, especially for datasets such as PURE and UBFC-rPPG. Future work could explore joint learning from heterogeneous supervision sources—direct respiratory belt signals, BVP-derived proxies, and unlabeled video—within a unified probabilistic framework that explicitly models label uncertainty. Ultimately, this investigation shows promising in-domain performance on COHFACE, as well as potential for generalization to other applications, but before it can be used clinically there will need to be validation on large numbers of diverse individuals including patients in hospital environments that may have challenging lighting conditions, different types of clothing worn by patients, and patients moving about constantly. Future clinical evaluations in this type of environment will be crucial to continue making progress toward being able to perform non-contact respiratory monitoring in the real world.

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Ethics Checklist

Name of student:	Shuyi Zhou
Student number:	P2411669
Ethics training completed: <i>(please specify)</i>	COMP6298-231-24/25
Programme:	☐ BSc in Computing (Final Year Project) ☒ MSc in Big Data and IoT (Project Report) ☐ PhD in Computer Applied Technology (Thesis)
Module:	COMP6298
Title of project:	TriBranchNet: Multi-Region Adaptive Fusion for Video-Based Respiratory Rate Estimation with Semi-Supervised Training Support
Supervisor:	Yue sun

Instructions

Answer the following questions honestly. If your answer falls into a shaded box, you must provide additional information on how you will handle the related ethics issues on the last page of this form.

Project summary

This project proposes TriBranchNet, a novel multi-region fusion framework with semi-supervised contrastive learning for video-based respiratory rate (RR) estimation, addressing the limitations of single-region camera-based methods that suffer from low signal-to-noise ratio, motion artifacts, and environmental interference. TriBranchNet integrates complementary respiratory signals from three anatomical regions: facial rPPG (capturing skin microcirculation modulation), neck rPPG (leveraging strong vascular responses to intrathoracic pressure), and chest optical flow (quantifying thoracic mechanical motion via Farneback dense optical flow). A lightweight Cross-Attention fusion module is designed to dynamically adjust feature weights based on real-time signal quality, enabling adaptive switching to reliable sources when individual regions degrade. The Contrast-Phys+ frequency-domain contrastive learning framework is adapted to the respiration domain, supporting semi-supervised training with unlabeled videos to reduce reliance on scarce annotated data. The model is systematically evaluated on COHFACE (primary benchmark), PURE, UBFC-rPPG, AIR, and NBHR datasets, with ablation, semi-supervised, and cross-domain experiments validating its accuracy, robustness, and generalization. This work advances non-contact RR monitoring, offering a robust solution for clinical and daily health management with potential for real-world deployment in all-scenario, long-term continuous monitoring.

Participants (Consent procedures and possible harm)

		Yes	No	N/A
1	Does your project include human participants? If no, you can skip the rest of this section.	Yes		
2	Will any of the participants be from vulnerable groups, e.g. children under 18, people with communication difficulties, patients in hospital? If yes, please specify: _____		No	
3	Will you tell participants that their participation is voluntary?	Yes		
4	Will you obtain written consent for participation?	Yes		
5	Will you tell participants that they may withdraw from the research at any time and for any reasons without penalty?	Yes		
6	Is there any realistic risk that any participants could experience either physical or psychological distress or discomfort?		No	

Data Protection

7	Will any non-anonymised and/or personalized data be generated and/or stored?		No	
8	Will you have access to documents containing sensitive data about living individuals?		No	

Researcher Safety

		Yes	No	N/A
9	Will you be exposed to any risks greater than those encountered in their normal study/working life? E.g. operation of non-standard equipment or extended		No	
10	Will you be exposed to highly addictive or illegal activities (e.g. gambling, drugs) during the research?		No	

This research involves human participants for video-based respiratory rate data collection and dataset utilization, the sole identified ethics consideration related to human research participation. All ethical risks associated with participant involvement are mitigated through strict adherence to informed consent protocols, voluntary participation principles, and data anonymization standards, aligned with Macao Polytechnic University's academic integrity and research ethics guidelines, as well as international human research ethics norms (Declaration of Helsinki).

Informed & Voluntary Participation: All participants are provided with a clear, plain-language research information sheet detailing the study's purpose, procedures (video recording of facial/neck/chest regions for physiological signal extraction), data usage, and potential benefits/risks (no physical/psychological harm is anticipated). Participants are explicitly informed their involvement is entirely voluntary, with no coercion or penalty for refusal to participate.

Written Consent & Unrestricted Withdrawal: Prior to any data collection, written informed consent is obtained from all participants (or legal guardians for minor participants if applicable). Consent forms include a clear clause stating participants may withdraw from the study at any time, for any reason, with no negative consequences; all their data will be immediately deleted upon withdrawal request.

Data Anonymization & Protection: All video and physiological data collected/used (from public datasets COHFACE, PURE, UBFC-rPPG, AIR, NBHR) is fully anonymized—personal identifiers (e.g., names, facial features that enable direct identification, contact information) are removed or blurred prior to data processing and storage. No non-anonymised/personalized data is generated, stored, or accessed throughout the research, eliminating risks of data breach or unauthorized use of personal information.

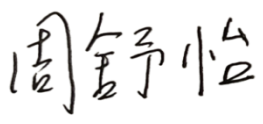
Minimization of Distress/Discomfort: The research involves only non-contact video recording (no physical sensors, invasive procedures, or stressful stimuli), and all recording sessions are conducted in a comfortable, private setting at a time convenient for participants. No realistic risk of physical or psychological distress/discomfort exists, and researchers are trained to pause/terminate recording if a participant reports any unease.

All public datasets used in this study are also vetted to confirm they comply with ethical research standards (i.e., collected with informed consent, fully anonymized, and made publicly available for academic research only), ensuring no secondary ethical risks from dataset utilization.

No other ethical issues (e.g., researcher safety, sensitive data access, vulnerable group misrepresentation) are identified in this research; all protocols will be documented and retained for academic audit as required by Macao Polytechnic University's research ethics regulations.

Declaration

Student	Supervisor
I have read the instruction carefully and reported honestly the potential ethics issues about the project.	To the best of my knowledge, the student has reported honestly the potential ethics issues about his/her project.

 Date: 2026/04/15	 Date: :2026/04/1
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This form is largely based on the ethical approval documents from Cardiff University <https://www.cs.cf.ac.uk/ethics/> and University of Glasgow <http://www.dcs.gla.ac.uk/ethics/>

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